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**HARMONISED EUROPEAN STANDARD**

**Satellite Earth Stations and Systems (SES);**
**Earth Stations on Trains (ESTs)**
**operating in the 11/12/14 GHz frequency bands;**
**Harmonised Standard for access to radio spectrum**

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REN/SES-00437

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# Foreword

This draft Harmonised European Standard (EN) has been produced by ETSI Technical Committee Satellite Earth Stations and Systems (SES), and is now submitted for the combined Public Enquiry and Vote phase of the ETSI Standardisation Request deliverable Approval Procedure (SRdAP).

The present document has been prepared under the Commission's standardisation request C(2015) 5376 final [i.5] to provide one voluntary means of conforming to the essential requirements of Directive 2014/53/EU on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment and repealing Directive 1999/5/EC [i.12].

Once the present document is cited in the Official Journal of the European Union under that Directive, compliance with the normative clauses of the present document given in table A.1 confers, within the limits of the scope of the present document, a presumption of conformity with the corresponding essential requirements of that Directive, and associated EFTA regulations.

| **Proposed national transposition dates** | |
|---|---|
| Date of latest announcement of this EN (doa): | 3 months after ETSI publication |
| Date of latest publication of new National Standard or endorsement of this EN (dop/e): | 6 months after doa |
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# Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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# 1 Scope

The present document specifies technical characteristics and methods of measurements for Earth Stations located on board Trains, which have the following characteristics.

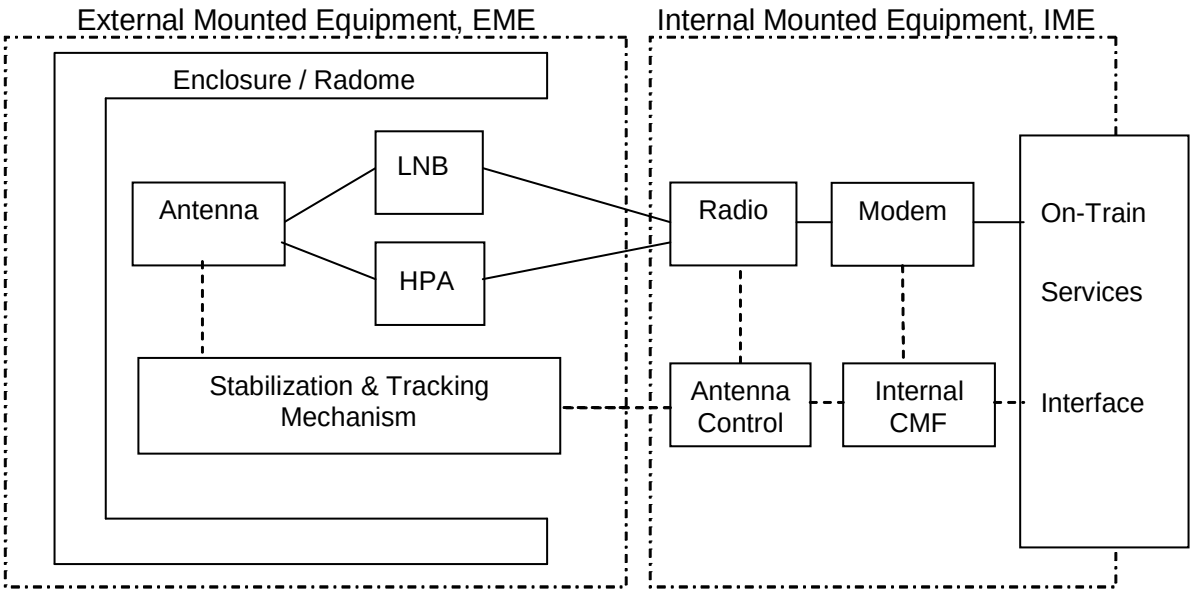



**Figure 1: EST System Overview**

- The EST may transmit and receive data when the train is in motion and also when the train is stationary.
- The EST operates in a railway environment and, therefore, may be subject to occasional disturbances and interruptions in the satellite link.
- The EST is operating as part of a satellite network (e.g. star, mesh or point-to-point) used for the distribution and/or exchange of information.
- The EST is comprised of all the equipment, electrical and mechanical, from the antenna itself to the interface with other communications equipment on a train (usually referred to as the terrestrial interface).
- The EST transmits on single carrier in the frequency range 14,00 GHz to 14,25 GHz, which is allocated to the Mobile Satellite Service (Earth-to-space), as secondary service.
- The EST receives in one or more frequencies within the range from 10,70 GHz to 12,75 GHz.
- The EST uses linear or circular polarization.
- The EST is designed to operate through a geostationary satellite (or a cluster of co-located geostationary satellites) that is at least 3° away from any other geostationary satellite operating in the same frequencies and over the same coverage area.
- The EST transmits at elevations greater than or equal to 7° relative to the local horizon.
- The EST is designed for unattended operation.
- The EST is controlled and monitored by a Network Control Facility (NCF). The NCF is outside the scope of the present document.

The present document applies to the EST with its ancillary equipment and its various telecommunication ports, and when operated within the boundary limits of the operational environmental profile as defined for the intended use of the EST and when installed as required by the defined intended use of the EST or in the user documentation.

NOTE: The relationship between the present document and essential requirements of article 3.2 of Directive 2014/53 [i.12] is given in annex A.

# 2 References

## 2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are necessary for the application of the present document.

[1] Void.

[2] Void.

[3] CISPR 16-1-5:2014+AMD1:2016: "Specification for radio disturbance and immunity measuring apparatus and methods - Part 1-5: Radio disturbance and immunity measuring apparatus - Antenna calibration sites and reference test sites for 5 MHz to 18 GHz".

## 2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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[i.1] Void.

[i.2] ECC Decision (20)02: "Harmonised use of the paired frequency bands 874.4- 880.0 MHz and 919.4-925.0 MHz and of the unpaired frequency band 1900-1910 MHz for Railway Mobile Radio (RMR)".

[i.3] Void.

[i.4] ETSI TR 102 215 (2004): "Electromagnetic compatibility and Radio spectrum Matters (ERM); Recommended approach, and possible limits for measurement uncertainty for the measurement of radiated electromagnetic fields above 1 GHz".

[i.5] Commission Implementing Decision C(2015) 5376 final of 4.8.2015 on a standardisation request to the European Committee for Electrotechnical Standardisation and to the European Telecommunications Standards Institute as regards radio equipment in support of Directive 2014/53/EU of the European Parliament and of the Council.

[i.6] EN 50155: "Railway applications - Electronic equipment used on rolling stock", (produced by CEN).

[i.7] BS EN 60068 Parts 2-6, 2-57, 2-65: "Environmental testing. Test methods for vibration and shock", (produced by CEN).

[i.8] EN 61373: "Railway applications. Rolling stock equipment. Shock and vibration tests", (produced by CEN).

[i.9] EN 60945: "Maritime navigation and radiocommunication equipment and systems - General requirements - Methods of testing and required test results", (produced by CEN).

[i.10] EN 55022: "Limits and methods of measurement of radio disturbance characteristics of information technology equipment", (produced by CEN).

[i.11] ETSI TR 102 375 (2005): "Satellite Earth Stations and Systems (SES); Guidelines for determining the parts of satellite earth station antenna radiation patterns concerned by the geostationary satellite orbit protection".

[i.12] Directive 2014/53/EU of the European Parliament and of the Council of 16 April 2014 on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment and repealing Directive 1999/5/EC (RE Directive).

[i.13] ETSI EG 203 336: "Guide for the selection of technical parameters for the production of Harmonised Standards covering article 3.1(b) and article 3.2 of Directive 2014/53/EU".

# 3 Definition of terms, symbols and abbreviations

## 3.1 Terms

For the purposes of the present document, the terms given in Directive 2014/53/EU [i.12] and the following apply:

**ancillary equipment:** equipment used in connection with an EST is considered as ancillary if the three following conditions are met:

a) the equipment is intended for use in conjunction with the EST to provide additional operational and/or control features (e.g. to extend control to another position or location); and

b) the equipment cannot be used on a stand alone basis, to provide user functions independently of the EST; and

c) the absence of the equipment does not inhibit the operation of the EST

**carrier-off radio state:** radio state in which the EST may transmit and does not transmit any carrier

NOTE 1: The phrase "the EST may transmit" means that all the conditions for transmission are satisfied (e.g. in a state where transmissions are permitted, no failure detected, and the EST is correctly pointed towards the satellite).

NOTE 2: The existence of a "Carrier-off" radio state depends on the system of transmission used. For ESTs designed for continuous transmission mode there may be no "Carrier-off" state.

**carrier-on radio state:** radio state in which the EST may transmit and transmits a carrier

**Control Channel (CC):** channel or channels by which ESTs receive control information from the NCF

**$EIRP_{max}$:** maximum e.i.r.p. capability of the EST as defined for the intended use of the EST

**emissions disabled radio state:** radio state in which the EST does not emit

NOTE: Examples of cases where the EST is in this radio state: before system monitoring pass, before the control channel is received, when a failure is detected, when an EST is commanded to disable, and when the EST is in a location requiring cessation of emissions.

**external control channel:** control channel which is either (i) carried by the EST network via the same or another satellite, but not within the internal protocol of the EST system, or (ii) carried by any other radio communication system

**external response channel:** response channel which is either (i) carried by the EST network via the same or another satellite, but not within the internal protocol of the EST system, or (ii) carried by any other radio communication system

**Externally Mounted Equipment (EME):** that part of the EST intended to be installed on the outside of the train (usually the roof), as defined for the intended use of the EST, or as indicated in the user documentation

NOTE 1: The EME unit is usually comprised of the following main parts:

a) The antenna sub-system which converts the incident radiation field into a guided wave and vice versa.

b) The Low Noise Block (LNB) down converter, which is a device that amplifies, with very low internal noise, the received signals in the Radio Frequency (RF) band and converts them to intermediate frequencies.

c) The up-converter and the power amplifier which convert from the intermediate frequency to RF and amplify the low level RF signals for transmission through the antenna subsystem.

d) The stabilization and tracking mechanics that ensure pointing of the antenna main beam towards the satellite within the required accuracy.

NOTE 2: The installation equipment (means of attachment) is outside the scope of the present document. However, the antenna structures and other components directly mounted on the antenna and forming an integral part of it, are subject to the specifications of the present document.

NOTE 3: Certain configurations may include more than one EME per EST. As for example, the EST may have separate transmit and receive antennas or it may have redundant transmit/receive antenna units.

**integral antenna:** antenna which may not be removed during the tests according to the EUT documentation

**internal control channel:** control channel which is carried by the EST network via the same satellite as used for transmission of user data and within the internal protocol structure of the EST system

**internal response channel:** response channel which is carried by the EST network via the same satellite as used for transmission of user data and within the internal protocol structure of the EST system

**Internally Mounted Equipment (IME):** part of the EST equipment which is installed inside the train and its connection cables with the EME

NOTE: The IME is usually comprised of:

a) the modem and the IF radio equipment;

b) the control logic, including that for the internal control and monitoring subsystem and the antenna tracking subsystem; and

c) the interfaces to equipment and services onboard the train.

**maximum relative wind speed:** addition of the magnitudes of the maximum wind speed and the maximum train velocity

**Network Control Facility (NCF):** set of functional entities that, at system level, monitor and control the correct operation of all ESTs in a network

**nominal antenna diameter:** diameter of the antenna defined for the intended use of the EST

NOTE 1: An antenna with a nominal antenna diameter D has the same technical characteristics as a circular aperture antenna of diameter D. For non circular aperture antenna, the nominal antenna diameter is defined for the intended use of the EST.

NOTE 2: Network operators might request antennas of a certain diameter. Then an antenna that is compliant with the requirement for nominal antenna diameter equal to the requested antenna diameter can be used.

**nominated bandwidth:** bandwidth of the EST radio frequency transmission nominated for the intended use of the EST. The nominated bandwidth is centred on the transmit frequency and does not exceed 5 times the occupied bandwidth

NOTE: The nominated bandwidth is wide enough to encompass all spectral elements of the transmission which have a level greater than the specified spurious radiation limits. The nominated bandwidth is wide enough to take account of the transmit carrier frequency stability. This definition is chosen to allow flexibility regarding adjacent channel interference levels which will be taken into account by operational procedures depending on the exact transponder carrier assignment situation.

**off-axis angle:** angle between the direction of the axis of the antenna main beam and the considered direction

**removable antenna:** antenna which may be removed during the tests according to the EUT documentation

**Response Channel (RC):** channel by which EST transmit monitoring information to the NCF

**spurious radiation:** any radiation outside the nominated bandwidth

**transmission disabled state:** EST is in this state when it is not authorized by the NCF to transmit

**transmission enabled state:** EST is in this state when it is authorized by the NCF to transmit

**Wanted signal occupied Bandwidth (BW):**

a) for a digital modulation scheme: the width of the signal spectrum 10 dB below the maximum in-band power spectral density

## 3.2 Symbols

For the purposes of the present document, the following symbols apply:

| | |
|---|---|
| dBc | ratio expressed in decibels relative to the e.i.r.p. of the unmodulated carrier |
| dBi | ratio of an antenna gain to the gain of an isotropic antenna, expressed in decibels |
| dBsd | Ratio expressed in decibels relative to the spectral density |
| dBW | ratio of a power to 1 watt, expressed in decibels |
| dBpW | ratio of a power to 1 picowatt, expressed in decibels |
| dBµV/m | ratio of an electric field to 1 µV/m, expressed in decibels (20 log(electric field /1 µV/m)) |

## 3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

| | |
|---|---|
| BW | Bandwidth |
| CC | Control Channel |
| CCF | Control Channel reception Failure |
| CCR | Control Channel correctly Received |
| CENR | Cessation of Emissions Not Requested |
| CEPT | Conférence Européenne des Postes et Télécommunications (European Conference of Postal and Telecommunications Administrations) |
| CER | Cessation of Emissions Requested |
| CISPR | Comité International Spécial des Perturbations Radioélectriques (International Special Committee on Radio Interference) |
| CMF | Control and Monitoring Functions |
| CW | Continuous Wave |
| EIRP | Equivalent Isotropically Radiated Power |
| EME | Externally Mounted Equipment |
| EN | European Standard |
| EST | Earth Station on board a Train |
| EUT | Equipment Under Test |
| FEC | Forward Error Correction |
| FS | Fixed Service |
| FSS | Fixed Satellite Service |
| GSM-R | Global System for Mobile communications - Railway |

| | |
|---|---|
| GSO | Geostationary Satellite Orbit |
| IME | Internally Mounted Equipment |
| LNB | Low Noise Block down converter |
| LO | Local Oscillator |
| NCF | Network Control Facility |
| OATS | Open Area Test Site |
| RC | Response Channel |
| RF | Radio Frequency |
| SMF | System Monitoring Fail |
| SMP | System Monitoring Pass |
| SNR | Signal to Noise Ratio |
| STE | Special Test Equipment |
| TDMA | Time Division Multiple Access |
| TxD | Transmission Disable command |
| TxE | Transmission Enable command |

# 4 Technical requirements specifications

## 4.1 General

### 4.1.1 Environmental profile

The technical requirements of the present document apply under the environmental profile for operation of the equipment, which shall be in accordance with its intended use, but as a minimum, shall be that specified in the test conditions contained in the present document. The equipment shall comply with all the technical requirements of the present document at all times when operating within the boundary limits of the operational environmental profile defined by its intended use.

The environmental profile shall include, the conditions regarding the train velocity, tangential and longitudinal acceleration, temperature range, damp heat, dry heat, relative humidity stress, shock and vibration, antenna pressure pulses and pressure gradients.

The equipment shall comply with the performance requirements of the present document under all operational environmental conditions defined by its intended use.

NOTE: For guidance on the range of values for parameters in the environmental profile, consultation to the following standards: EN 50155 [i.6]; BS EN 60068 [i.7]; EN 61373 [i.8] is recommended. For certain environmental parameters not covered in these standards, it may be useful to consult EN 60945 [i.9].

### 4.1.2 Operational configurations

Under operational conditions an EST may dynamically change the occupied bandwidth and other transmission parameters (e.g. FEC, modulation, symbol rate) of the transmitted signal. For each occupied bandwidth an $EIRP_{max}$ and a nominated bandwidth shall be provided with the documentation of the EUT. For the purposes of verifying that the EST complies with the specifications in the present document, the worst case combination of transmission parameters shall be identified. However, the specifications in the present document apply to the EST for each occupied bandwidth and other transmission parameters.

### 4.1.3 EST states and radio states

For the purpose of the present document the following four EST states are defined, without presuming the effective implementation of the EST state machine:

a) "Non valid";

b) "Initial phase";

c) "Transmission disabled"; and

d) "Transmission enabled".

The four EST states are represented on figure 2 and are used in clause 4.2.8 for the specification of the Control and Monitoring Functions (CMFs).

In the "Non-valid" state and in the "Transmission disabled" state the EST is not allowed to transmit. In the "Transmission-enabled" state the EST is allowed to transmit. In the "Initial phase" state the EST is only allowed to transmit initial bursts or is waiting for a transmit enable/disable command.

The "Initial phase" is divided into three substates:

a) "Initial phase-Standby" prior to the transmission of the first initial burst or when no initial bursts are transmitted;

b) "Initial phase-BurstOn" during the transmission of the initial bursts;

c) "Initial phase-BurstOff" between initial bursts.

NOTE: ESTs which do not transmit initial bursts have no "Initial phase-BurstOn" state and no "Initial phase-BurstOff" state.

The EST can transmit when all the conditions for transmission are satisfied (e.g. in a state where transmissions are permitted, no failure detected, correctly pointed towards the satellite and there is no requirement for cessation of emissions).

The following radio states of the EST are defined:

a) "Emissions disabled" when the EST does not transmit any carrier;

b) "Carrier-off" when the EST may transmit and does not transmit any carrier;

c) "Carrier-on" when the EST may transmit and transmits a carrier.

Table 1a gives the only possible combinations of the EST states and radio states which shall apply, with some examples of associated events.

When the EST transmits several carriers having different frequencies, an EST state machine as described above may be associated with each carrier or each set of carriers.

**Table 1a: EST states and radio states of the EST**

| EST states and substates | Radio states | Examples of events |
|---|---|---|
| "Non valid" | "Emissions disabled" | After-power on, or<br>After any failure, or<br>During the checking phase |
| "Initial phase" | | When waiting for a transmission enable or disable command from the NCF |
| "Initial phase-Standby" | "Emissions disabled" | Before the first initial burst transmissions; or<br>In locations where no transmission is allowed |
| "Initial phase-BurstOn" | "Carrier-on" | During the transmission of each initial burst, and the pointing is correct |
| "Initial phase-BurstOff" | "Carrier-off" | Between initial bursts, or<br>When the pointing threshold is exceeded |
| "Transmission enabled" | "Carrier-off" | When no carrier is transmitted, or<br>When receive synchronization is lost, or<br>When the pointing threshold is exceeded |
| | "Carrier-on" | During transmission of carrier(s), and the pointing is correct |
| "Transmission disabled" | "Emissions disabled" | When a disable command from the NCF has been received and waiting for a transmission enable command from the NCF, or<br>In locations where no transmission is allowed |

# 4.2 Conformance requirements

## 4.2.1 Off-axis spurious radiation

### 4.2.1.1 Justification

To limit the level of interference to terrestrial and satellite radio services to meet the essential requirements as referred to in the scope of the present document.

### 4.2.1.2 Specification

The following specifications apply to the EST transmitting at equivalent isotropically radiated power (e.i.r.p.) values up to and including $EIRP_{max}$.

1) The electric field strength level of any radiation from the EST in the frequency range from 30 MHz to 1 GHz shall not exceed the limits specified in table 1b.

**Table 1b: Limits of radiated electric field strength**

| Frequency range | Limit (Quasi Peak) | Limit (Peak) | Measurement bandwidth | Measuring distance |
|---|---|---|---|---|
| 30 MHz to 230 MHz (see note 1) | 40 dBµV/m | no limit applies | 120 kHz | 10 m |
| 230 MHz to 1 GHz (see note 1) (see note 3) | 47 dBµV/m | no limit applies | 120 kHz | 10 m |
| 876 MHz to 880 MHz (see note 2) | 22 dBµV/m | no limit applies | 120 kHz | 3 m |
| 921 MHz to 925 MHz (see note 2) | 22 dBµV/m | no limit applies | 120 kHz | 3 m |
| NOTE 1: The limits in these rows are taken from EN 55022 [i.10]. NOTE 2: The frequency ranges 876 MHz to 880 MHz and 921 MHz to 925 MHz are the base and mobile receive bands respectively of GSM-R (see ECC Decision (20)02 [i.2]). NOTE 3: Except in the band from 876 MHz to 880 MHz and from 921 MHz to 925 MHz as specified in this table. | | | | |

The Peak and Quasi Peak detectors shall be in accordance with CISPR 16-1-5 [3].

2) When the EST is in the "Emissions disabled" radio state, the off-axis spurious e.i.r.p. from the EST, in any 100 kHz band, shall not exceed the limits in table 2, for all off-axis angles greater than 7° or greater than the minimum elevation angle defined for the intended use of the EST, whichever is lower.

**Table 2: Limits of spurious e.i.r.p. - "Emissions disabled" radio state**

| Frequency band | e.i.r.p. limit (dBpW) | Measurement bandwidth |
|---|---|---|
| 1,0 GHz to 10,7 GHz | 48 | 100 kHz |
| 10,7 GHz to 21,2 GHz | 54 | 100 kHz |
| 21,2 GHz to 40,0 GHz | 60 | 100 kHz |

The lower limits shall apply at the transition frequency.

3) This specification applies outside the nominated bandwidth. In the "Carrier-on" and "Carrier-off" radio states, the off-axis spurious e.i.r.p. density from the EST, shall not exceed the limits in table 3, for all off-axis angles greater than 7° or greater than the minimum elevation angle defined for the intended use of the EST, whichever is lower.

**Table 3: Limits of spurious e.i.r.p. - "Carrier-on" and "Carrier-off" radio states**

| Frequency band | e.i.r.p. limit (dBpW) | Measurement bandwidth | Remarks |
|---|---|---|---|
| 1,0 GHz to 3,4 GHz | 49 | 100 kHz | |
| 3,4 GHz to 10,7 GHz | 55 | 100 kHz | |
| 10,7 GHz to 13,75 GHz | 61 | 100 kHz | |
| 13,75 GHz to 14,0 GHz | 95 | 10 MHz | See note 1 |
| 14,00 GHz to 14,25 GHz | | | See note 2 |
| 14,250 GHz to 14,75 GHz | 95 | 10 MHz | See note 1 |
| 14,75 GHz to 21,2 GHz | 61 | 100 kHz | |
| 21,2 GHz to 40,0 GHz | 67 | 100 kHz | |
| NOTE 1: This limit may be exceeded in a frequency band which shall not exceed 50 MHz, centred on the carrier frequency, provided that the on-axis e.i.r.p. density at the considered frequency is 50 dB below the maximum on-axis e.i.r.p. density of the signal (within the nominated bandwidth) expressed in dBW/100 kHz.<br>NOTE 2: A limit may be required in regions outside CEPT where FS systems have an allocation and are operating in the band 14,00 GHz to 14,25 GHz. This requirement is outside the scope of the present document. | | | |

The lower limits shall apply at the transition frequency.

In the frequency band 28,0 GHz to 29,0 GHz, for any 20 MHz band within which one or more spurious signals exceeding the above limit of 67 dBpW/100 kHz are present, then the power of each of those spurious signals exceeding the limit shall be added in watts, and the sum shall not exceed 78 dBpW.

4) These limits are applicable to the complete EST equipment, comprising the IME and EME including cabling between the units.

#### 4.2.1.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.2.

### 4.2.2 On-axis spurious radiation

#### 4.2.2.1 Justification

To limit the level of interference to satellite radio services.

#### 4.2.2.2 Specification

##### 4.2.2.2.1 "Carrier-on" radio state

The following specification applies to the EST transmitting at e.i.r.p. values up to $EIRP_{max}$.

In the 14,0 GHz to 14,25 GHz band the e.i.r.p. spectral density of the spurious radiation outside a bandwidth of 5 times the occupied bandwidth centred on the carrier centre frequency shall not exceed 4 - K dBW in any 100 kHz band.

The e.i.r.p. spectral density of the spurious radiation outside the nominated bandwidth and within a bandwidth of 5 times the occupied bandwidth centred on the carrier centre frequency, shall not exceed 18 - K dBW in any 100 kHz band.

The factor K accounts for a reduction on the on-axis spurious radiation level in case of multiple ESTs operating on the same frequency, and the value is given by one of the following cases:

1) For the case where only one EST transmits at any one time on a given carrier frequency, the value of K is 0.

2) For the case where several ESTs are expected to transmit simultaneously on a given carrier frequency at the same e.i.r.p., the value of K = 10 log (N) where N is the maximum number of these ESTs. The value of N shall not be exceeded by the number of ESTs actually transmitting simultaneously on a given carrier frequency at the same e.i.r.p. for more than 0,01 % of the time during at least 1 minute. The value of N and the operational conditions of the system shall be defined for the intended use of the EST.

3) For the case where several ESTs are expected to transmit simultaneously on a given carrier frequency at different e.i.r.p. levels, the value of K = 10 log ($EIRP_{Aggregate}$/EIRP), where:

   - EIRP is the on-axis e.i.r.p. (in Watt) of the EST within the nominated bandwidth; and
   - $EIRP_{Aggregate}$ is the expected maximum on-axis aggregate e.i.r.p. (in Watt) within the nominated bandwidth of the EST system towards the satellite.

   The value of $EIRP_{Aggregate}$ shall not be exceeded by the actual maximum on-axis aggregate e.i.r.p. within the nominated bandwidth of the EST system towards the satellite for more than 0,01 % of the time during at least 1 minute.

The value of $EIRP_{Aggregate}$ and the operational conditions of the EST network shall be defined for the intended use of the EST.

NOTE 1: The on-axis spurious radiations, outside the 14,0 GHz to 14,25 GHz band, are indirectly limited by clause 4.2.1.2. Consequently no specification is needed.

NOTE 2: Intermodulation limits inside the band 14,0 GHz to 14,25 GHz are to be determined by system design and are subject to satellite operator specifications.

For ESTs designed to transmit several carriers on different frequencies simultaneously (multicarrier operation), the above limits only apply to each individual carrier when transmitted alone.

#### 4.2.2.2.2 "Carrier-off" and "Emissions disabled" radio states

In the 14,0 GHz to 14,25 GHz band the e.i.r.p. spectral density of the spurious radiation outside the nominated bandwidth shall not exceed -21 dBW in any 100 kHz band.

### 4.2.2.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.3.

## 4.2.3 Off-axis e.i.r.p. emission density within the band

### 4.2.3.0 General

The present clause deals with the off-axis e.i.r.p. emission density (co-polarization and cross-polar) within the band 14,0 GHz to 14,25 GHz.

### 4.2.3.1 Justification

Protection of other satellite (uplink) systems.

### 4.2.3.2 Specification

The following specifications apply to the EST transmitting at e.i.r.p. values up to $EIRP_{max}$.

The maximum e.i.r.p. in any 40 kHz band within the nominated bandwidth of the co-polarized component in any direction $\phi$ degrees from the antenna main beam axis shall not exceed the following limits:

$$33 - 25 \log (\phi + \delta\phi) - H \text{ dBW} \quad \text{for} \quad 2{,}5° \leq \phi + \delta\phi \leq 7{,}0°;$$

$$+12 - H \text{ dBW} \quad \text{for} \quad 7{,}0° < \phi + \delta\phi \leq 9{,}2°;$$

$$36 - 25 \log (\phi + \delta\phi) - H \text{ dBW} \quad \text{for} \quad 9{,}2° < \phi + \delta\phi \leq 48°;$$

$$- 6 - H \text{ dBW} \quad \text{for} \quad \phi + \delta\phi > 48°;$$

where:

- $\phi$ is the angle, in degrees, between the main beam axis and the direction considered; and
- $\delta\phi$ is the pointing error threshold, in degrees, as defined for the intended use of the EST (see clause 4.2.5.2).

For ESTs designed to transmit always at $EIRP_{max}$, H (in dB) is the maximum number of ESTs which may transmit at $EIRP_{max}$ in the same carrier frequency band as defined for the intended use of the EST.

For ESTs designed to operate in an FSS network where the e.i.r.p. of each EST is determined by the NCF and where the NCF is in charge of the compliance of the aggregate e.i.r.p. density with the above mask, H is the margin as defined for the intended use of the EST for compliance with the mask when the EST is transmitting at $EIRP_{max}$. In the case where the NCF uses the antenna pattern or the off-axis $EIRP_{sd}$ the applicable pattern shall be defined for the intended use of the EST, the value of H shall be set to 0 dB and the EST e.i.r.p. density shall not exceed the e.i.r.p. density corresponding to the defined pattern. This margin H or this pattern may be a function of the position of the EST relative to the GSO arc (e.g. in the case of active antennas).

For $\phi > 70°$ the values given above can be increased to 4 - H dBW

In addition the maximum e.i.r.p. in any 40 kHz band within the nominated bandwidth of the cross-polarized component in any direction $\phi$ degrees from the antenna main beam axis shall not exceed the following limits:

$$23 - 25 \log (\phi + \delta\phi) - H \text{ dBW} \quad \text{for} \quad 2{,}5° \leq \phi + \delta\phi \leq 7{,}0°;$$

$$+2 - H \text{ dBW} \quad \text{for} \quad 7{,}0° < \phi + \delta\phi \leq 9{,}2°;$$

where $\phi$, and H are as defined above.

For non-continuous transmission, the above limits may not apply for a specific portion of each burst as defined for the intended use of the EST. This excluded portion shall not exceed 50 µsec or 10 % of the burst, whichever is the smaller.

The excluded portion shall have characteristics similar to the remaining part of the burst:

a) same symbol rate and modulation; and

b) same or lower maximum amplitude.

NOTE 1: The specification above is intended to take account of the spectrum of the preamble of bursts in TDMA systems.

NOTE 2: The satellite operator may require lower off-axis e.i.r.p. limits in compliance with the intersystem coordination agreements. In all cases, off-axis e.i.r.p. emissions are subject to compliance with national regulations and the relevant intersystem coordination agreements.

Any antenna off-axis direction may be defined by a pair of values ($\alpha$, $\varphi$) where $\varphi$ is the off-axis angle of that direction with the antenna main beam axis and $\alpha$ is the angle of the plane defined by that direction and the antenna main beam axis with any arbitrary plane containing the antenna main beam axis.
The range of values of $\varphi$ and $\alpha$ is from $\varphi_{min} - \delta\varphi$ to 180° for $\varphi$, and from -180° to +180° for $\alpha$.

The above limits only apply to any off-axis direction ($\alpha$, $\varphi$) within ± 3° of the visible part of the GSO. The concerned directions shall be any direction within the ($\alpha$, $\varphi$) domain unless it can be demonstrated by documentary evidence that only a limited subset of the ($\alpha$, $\varphi$) domain is concerned.

The determination of the (α, φ) subset shall take into account the operational conditions for which the EST is designed, as defined for the intended use of the EST or indicated as par of the data provided with the EST. These conditions shall include:

a) the range of latitudes of the EST;

b) the minimum elevation pointing angle;

c) the type of antenna mount (e.g. with azimuth and elevation axes or equatorial);

d) the maximum static and dynamic alignment errors of the antenna mount axes;

e) the maximum static and dynamic alignment errors of the antenna major axis with respect to the GSO arc;

f) the range of adjustment for the major axis of the antenna for antennas with asymmetric main beam;

g) the direction of the electric field radiated by the satellite with respect to the Earth's axis, when the electric field is used for the antenna alignment.

The alignment errors shall not exceed the maximum values defined for the intended use of the EST when applying the alignment method defined for the intended use of the EST or as the data provided with the EST.

NOTE 3: ETSI TR 102 375 [i.11] gives guidance for the determination of the concerned subset within the (α, φ) domain.

#### 4.2.3.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.4 with the test method as in clauses 6.4.0 and 6.4.1 and with the results being computed in accordance with clause 6.4.2.

### 4.2.4 Carrier suppression

#### 4.2.4.1 Justification

To allow for the satisfactory suppression of transmissions of the EST by the NCF, under any fault condition and under any cessation of emissions condition.

#### 4.2.4.2 Specification

In the "Carrier-off" and in the "Emissions disabled" radio states the on-axis e.i.r.p. shall not exceed 4 dBW in any 100 kHz band within the nominated bandwidth.

#### 4.2.4.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.5.

### 4.2.5 Antenna characteristics

#### 4.2.5.1 Justification

Protection of transmissions to adjacent satellites and cross-polarized transponders on the same satellite.

#### 4.2.5.2 Specification

a) Mechanical stability:

The maximum values within which the EST is designed to operate in accordance with clause 4.1.1 for the following parameters shall be defined for the intended use:

a) maximum train speed;

b) maximum wind speed, which shall not be less than 100 km/h;

c) maximum pressure pulse, which shall not be less than 6 kPa (Peak to peak for 3 seconds); and

d) maximum pressure gradient, which shall not be less than 100 kPa/s.

The EST shall not show any sign of permanent distortion that could affect the radiation subsystem after the application of the load under the maximum values defined for the intended use of the EST for parameters above, including the Maximum relative wind speed.

b) Pointing error detection:

The EST shall be able to detect the pointing error. The EST, when in the "Carrier-on" radio state, shall enter the "Carrier-off" radio state when the instantaneous pointing error exceeds the pointing error threshold, $\delta\phi$, relative to the direction of the wanted satellite at its actual position, within T seconds. The values of $\delta\phi$ and T shall be defined for the intended use of the EST and the value of T shall not exceed 5 seconds. The EST shall not re-enter the "Carrier-on" radio state until the instantaneous pointing error is within $\delta\phi$ for a period of $2 \times$ T seconds.

The maximum time ($IT_{max}$) that the EST can remain in "Transmission enabled" state and "Carrier off" radio state (see clause 4.2.8.8) shall be defined for the intended use of the EST. If the pointing error threshold is exceeded for more than $IT_{max}$ then the EST shall enter the "Initial phase" state. The EUT shall correctly detect that the induced pointing error has exceeded the pointing error threshold and enters the "Carrier off" radio state no later than T seconds after the pointing error threshold is exceeded, where T is the response time defined for the intended use of the EST.

The ranges of values for $\delta\phi$ as functions of on-axis e.i.r.p. spectral density such that the e.i.r.p. density limits in clause 4.2.3.2 are not exceeded shall be defined for the intended use of the EST.

#### 4.2.5.3 Conformance tests

Conformance tests shall be carried out in accordance with clauses 6.6 to 6.8. In addition, clause 6.11.8 contains the conformance test for inhibition of transmission when the outage time exceeds $IT_{max}$.

### 4.2.6 Cessation of emissions of the EST

#### 4.2.6.1 Justification

For cessation of emissions of the EST where the EST is not allowed to transmit.

#### 4.2.6.2 Specification

##### 4.2.6.2.1 Specification 1: Mode of cessation of emissions

At least one of the following two modes of cessation of emissions shall be implemented:

a) the NCF determines that the EST shall cease emissions; and/or

b) the EST autonomously determines that it shall cease emissions.

In addition, there shall be a "single-action" means (e.g. operating a switch) by which a local operator may disable the EST and thereby cease emissions.

The EST interfaces involved in the cessation of emissions consistent with the intended use of the EST shall be provided with the documentation of the EUT including:

a) the list of relevant parameters which are collected by the EST or the NCF for determination as to whether the EST should cease emissions;

b) the list of these relevant parameters which are used by the EST;

c) the list of these relevant parameters which are transmitted by the EST to the NCF;

d) the list of the relevant parameters which are received by the EST from the NCF;

e) for the collected relevant parameters, the EST interface (s), including the protocols, the timing, the ranges of the values, the speed of the variations and the required accuracies;

f) for the relevant parameters transmitted to the NCF, the EST interface with the NCF, including the protocols and the timing;

g) for the transmission parameter received from the NCF, the EST interface with the NCF, including the protocols and the timing.

#### 4.2.6.2.2 Specification 2: Conditions for cessation of emissions

For EST networks where the NCF determines completely or partially where and under which conditions the emissions of the EST shall be ceased, the collection of the relevant parameters by the EST and the exchange of information between the EST and the NCF shall be sufficient for the NCF to determine when to cease emissions with the accuracy defined for the intended use of the EST and to inform in time the EST to cease emissions.

For ESTs which determine completely where and under which conditions the emissions shall be ceased, the collection of the relevant parameters by the EST shall be sufficient for the EST to determine when to cease emissions with the accuracy defined for the intended use of the EST.

The determination of these conditions shall take into account the inaccuracy of the EST location and of the boundaries of the authorized operating areas in the data base used either by the EST or the NCF, as defined for the intended use of the EST.

The conditions for cessation of emissions shall take into account at least the following parameters:

a) the location of the EST and the boundaries of the authorized operating area;

b) the operating parameters specified in clauses 4.2.1 through 4.2.6;

c) the carrier frequency and the authorized frequency bands.

#### 4.2.6.2.3 Specification 3: Cessation of emissions

A condition requiring cessation of emissions occurs either when the EST receives the command from the NCF at its input or when the EST determines autonomously on the need to cease emissions.

When in the "Transmission enabled" state a condition requiring cessation of emissions occurs, the EST shall cease transmissions and enter the "Initial Phase" state.

When in the "Transmission disabled" state, the EST shall not leave that state for the "Transmission enable" state as long as a condition requiring cessation of emissions exists or the last CC command received from the NCF is a transmission disable command.

When in the "Initial phase-BurstOn" or "Initial phase-BurstOff" substate a condition requiring cessation of emissions occurs, the EST shall cease transmissions and enter the "Initial phase-Standby" substate.

When in the "Initial phase-Standby" substate, the EST shall not leave that state for the "Initial phase-BurstOn" or the "Transmission enabled" states as long as a condition requiring cessation of emissions exists.

The time for transition in any state from the occurrence of a condition requiring cessation of emissions to the "Emissions disabled" radio state shall not exceed 1 second.

#### 4.2.6.2.4 Specification 4: Fault conditions

Any collection of the relevant parameters by the EST or transmission of these parameters to the NCF, which have not been completed correctly within the required delay(s) as defined for the intended use of the EST, shall be considered as a fault condition.

Any transmission parameter not received or not correctly received from the NCF within the required delay defined for the intended use of the EST shall be considered as a fault condition.

These fault conditions shall be processed as conditions requiring cessation of emissions.

#### 4.2.6.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.9.

### 4.2.7 Identification of EST

#### 4.2.7.1 Justification

To fulfil the requirements for network management and for potential use by duly authorized entities.

#### 4.2.7.2 Specification

The EST shall be designed such that it is possible for the network operator to identify which ESTs are transmitting with at least 100 m resolution in a given geographic area. The means by which identification of ESTs is accomplished shall be provided with the documentation of the EUT. In the case where an external system is required, additional means which are necessary for identification of the EST shall be provided with the documentation of the EUT.

#### 4.2.7.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.10.

### 4.2.8 Control and Monitoring Functions (CMFs)

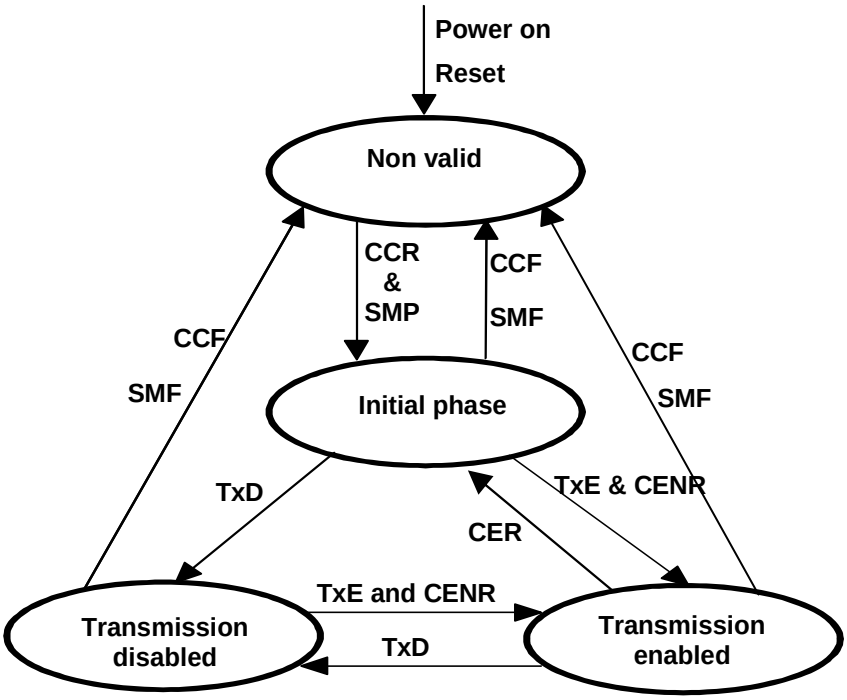
#### 4.2.8.1 CMF state diagram

The following minimum set of CMFs shall be implemented in ESTs in order to minimize the probability that they originate unwanted transmissions that may give rise to harmful interference to other systems.

In the "Non-valid" state and in the "Transmission disabled" state the EST shall not transmit. In the "Transmission-enabled" state the EST is allowed to transmit. In the "Initial phase" state the EST is only allowed to transmit initial bursts.

NOTE 1: The restrictions in the "Initial phase" state are for the protection of other systems when the EST is entering the system after a power-on or a reset. These initial burst restrictions do not apply to the EST transmissions in the "Transmission-enabled" state and once a transmission enable command has been received by the EST the EST may transmit or not transmit as required.

Under any fault condition the EST shall enter the "Non-valid" state where the e.i.r.p. limits for the "Emissions disabled" radio state apply.




CER: Cessation of Emissions Required;
CENR: Cessation of Emissions Not Required;
SMP: System Monitoring Pass;
SMF: System Monitoring Fail;
TxE: Transmission Enable command;
TxD: Transmission Disable command;
CCR: Control Channel correctly Received;
CCF: Control Channel reception Failure.

**Figure 2: State transition diagram of the control and monitoring function of an EST**

NOTE 2: From "Transmission disabled" state a TxE command may also result in a transition towards the "Initial phase" state.

When the EST transmits several carriers having different frequencies, an EST state machine as described above may be associated with each carrier or each set of carriers. The events then apply to the subsystem associated with the specific carrier or the specific set of carriers, rather than the whole EST.

### 4.2.8.2 Processor monitoring

#### 4.2.8.2.1 Justification

To ensure that the EST can suppress its transmissions in the event of a processor sub-system failure.

#### 4.2.8.2.2 Specification

The EST shall incorporate a processor monitoring function for each of its processors involved in maintaining the performance requirements of the present document and in Control and Monitoring Functions (CMF).

The processor monitoring function shall detect failures of these processors' hardware and software.

In the "Transmission enabled" state, the EST shall enter the "Non valid" state or the "Carrier-off" radio state no later than 1 second after any fault condition occurs, and in any case it shall enter the "Non valid" state within 30 seconds until the processor monitoring function has determined that all fault conditions have been cleared. In any other state after any fault condition occurs, the EST shall enter the "Non valid" state within 30 seconds until the processor monitoring function has determined that all fault conditions have been cleared.

#### 4.2.8.2.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.11.2.

### 4.2.8.3 Transmit subsystem monitoring

#### 4.2.8.3.1 Justification

To ensure the inhibition of transmissions that are potentially harmful to other systems in the event of incorrect operation of the transmit frequency generation sub-system.

#### 4.2.8.3.2 Specification

The EST shall monitor the operation of its transmit frequency generation sub-system and shall be able to detect:

a) loss of frequency lock;

b) absence of Local Oscillator (LO) output signal.

No later than 1 second after any of these fault conditions of the transmit frequency generation sub-system occurs, the EST shall enter the "Non-valid" state until the transmit sub-system monitoring function has determined that all fault conditions have been cleared.

#### 4.2.8.3.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.11.3.

### 4.2.8.4 Power-on/Reset

#### 4.2.8.4.1 Justification

To demonstrate that the EST achieves a controlled non-transmitting state following the powering of the unit, or the occurrence of a reset made by a local operator when this function is implemented.

#### 4.2.8.4.2 Specification

Following a manual reset, when this function is implemented, the EST shall enter the "Non-valid" state.

During and following "power-on" the EST shall remain in the "Non-valid" state until all the conditions for entering "Initial phase" have been satisfied.

#### 4.2.8.4.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.11.4.

### 4.2.8.5 Control Channel (CC) and Response Channel (RC)

#### 4.2.8.5.1 Justification

Control Channels (CC) are used by ESTs to receive control information from the NCF and Response Channels (RCs) to respond to the NCF. For EST designed to operate within networks where the NCF determines that the EST shall cease emissions, then the CCs and RCs are also used for the dialogue with the NCF.

To ensure that the EST cannot transmit unless it correctly receives the CC messages from the NCF.

#### 4.2.8.5.2 Specification

##### 4.2.8.5.2.1 Specification 1: types of CCs and RCs

a) The EST shall have at least one CC from the NCF. If exchange of information with the NCF is necessary for operation of the EST then the EST shall have at least one RC to the NCF (see clause 4.2.6). Each CC and each RC may be internal or external.

The types (internal or external) of each CC and each RC shall be defined for the intended use of the EST.

NOTE 1: The availability of the network carrying the external CC(s) and RC(s) and the numbers of external CC(s) and RC(s) are not within the scope of the present document.

NOTE 2: Some satellite operators may require that internal CC(s) and or RC(s) are available.

b) The connection between the NCF and the EST via the CCs and RCs shall be either permanent or shall be set up on a call by call basis through a switched network. In case of connection through a switched network the EST shall be able to receive calls from the NCF and to initiate calls towards the NCF in order to set up the CCs and RCs.

c) The EST with an external CC shall not transmit without receiving an appropriate signal from the NCF through the satellite to which the EST transmits, indicating to the EST that the NCF is alive and insuring that the EST is pointing to the target satellite.

##### 4.2.8.5.2.2 Specification 2: CC Reception

a) The EST shall enter the "Non-valid" state immediately after a period not exceeding 30 seconds without correct reception of the CC from the NCF, i.e. without being able to receive CC messages from the NCF. This event is called a "CC disruption":

- In the case of external CCs and RCs without permanent connection of the NCF with the EST, the ability to receive CC messages from the NCF is the ability at any time to receive calls and messages within the timing requirements of the present document from the network through which is connected the NCF.

- The inability to receive CC messages from the NCF may be due to the following various causes but not limited to them: no received signal from the NCF or from the network, a too low level received signal, no network accessible, the inability to lock onto the received carrier frequency, to demodulate, to decode, to receive calls and/or messages, a hardware failure or power off.

b) The EST shall remain in the "Non-valid" state as long as the EST is unable to receive CC messages from the NCF.

c) From the "Non-valid" state the EST may enter the "Initial phase" state if the following conditions are met:

- the EST is able to receive CC messages from the NCF; and

- no fault conditions are present.

#### 4.2.8.5.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.11.5.

### 4.2.8.6 Network control commands

#### 4.2.8.6.1 Justification

These requirements ensure that the EST is capable of:

a) retaining a unique identification in the network;

b) receiving commands from the NCF through its CC(s) and executing those commands.

#### 4.2.8.6.2 Specification

The EST shall hold, in non-volatile memory, its unique identification code in the network.

The EST shall be capable of receiving through its CCs dedicated messages (addressed to the EST) from the NCF, and which contain:

a) Transmission Enable commands (TxE);

b) Transmission Disable commands (TxD).

When in the "Initial phase" or "Transmission enabled" states, once a transmission disable command is received, within 1 second the EST shall enter into, and shall remain in the "Transmission disabled" state until the transmission disable command is superseded by a subsequent transmission enable command (see also clause 4.2.6).

When in the "Initial phase" or "Transmission disabled" states, once a transmission enable command is received, the EST may enter into the "Transmission enabled" state.

When entering the "Initial phase" from the "Non-valid" state, the last TxE or TxD command received from the NCF may be used by the EST to enter the "Transmission enabled" state or the "Transmission disabled" state, respectively, if since the time of reception of that command no "CC disruption" occurred.

NOTE: The physical unit in charge of the reception of the CCs, of the transmissions of the RCs and of the CC commands reception may be a separate and independent unit from the other units of the EST and it may be common to several ESTs.

#### 4.2.8.6.3 Conformance test

Conformance tests shall be carried out in accordance with clause 6.11.6.

### 4.2.8.7 Initial burst transmission

#### 4.2.8.7.1 Justification

Restrictions on the initial burst transmissions are necessary to limit disturbance to other services.

#### 4.2.8.7.2 Specification

For systems where no transmission enable command is foreseen without request from the EST, in the "Initial phase" state the EST may transmit initial bursts:

a) The EST shall not start the transmission of initial bursts before having obtained a confirmation that no cessation of emission applies where the EST is located. This confirmation shall be obtained either by local means of determination of the cessation of emissions (see clause 4.2.6) or from the NCF via an external control channel.

b) The duty cycle of the burst retransmission shall not exceed 0,2 %, where the duty cycle is defined as the ratio of burst duration to the duration between two successive bursts.

c) Each burst shall not carry more than the minimum number of data bytes excluding the burst preambles and the FEC coding bits necessary to establish synchronization.

d) The initial burst shall be transmitted at an e.i.r.p. no greater than $EIRP_{max}$.

e) The duration between two successive bursts shall not be less than the required NCF response time as defined for the intended use of the EST. This response time is defined as the duration, measured at the EST, between an initial burst transmission and the reception and processing of a transmission enable or disable command from the NCF which is never exceeded during 99 % of the cases under normal conditions in the system for which the EST is designed, as defined for the intended use of the EST.

#### 4.2.8.7.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.11.7.

#### 4.2.8.8 Inhibition of transmissions

##### 4.2.8.8.1 Justification

To ensure the correct inhibition of transmissions that are potentially harmful to other systems and persons in the event of signal blockage.

##### 4.2.8.8.2 Specification

The EST shall inhibit transmissions whenever there is a loss of receive carrier (e.g. synchronization or signal level).

EST shall enter the "Carrier-off" radio state within a period not exceeding 0,5 seconds, when these conditions persist during that period, and shall remain in this radio state until the receiver carrier has been restored.

The maximum outage time ($IT_{max}$) for this condition ("Transmission enabled" state and "Carrier off" radio state) shall be provided with the documentation of the EUT. If the receive carrier is not restored within $IT_{max}$, then the EST shall transition to the "Initial phase" state.

The time to transition from "Carrier off" to "Carrier on" radio state ($IT_{trans}$) when in "Transmission enabled" state after the receive carrier is restored within $IT_{max}$ shall be provided with the documentation of the EUT.

##### 4.2.8.8.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.11.8.

### 4.2.9 Receive antenna off-axis gain pattern

#### 4.2.9.1 Justification

To protect the wanted signals from interference from terrestrial services and from other satellite services.

#### 4.2.9.2 Specification

The maximum antenna gain of each of the co-polarized components in any direction φ degrees from the antenna main beam axis shall not exceed the following limits:

| | | |
|---|---|---|
| $G = 32 - 25 \log \varphi$ | dBi | for $\varphi_{min} \leq \varphi < 48°$ |
| $G = -10$ | dBi | for $48° \leq \varphi \leq 85°$ |
| $G = 0$ | dBi | for $85° \leq \varphi \leq 180°$ |

where:

$\varphi_{min}$ = 1° or 100 λ/*D* degrees, whichever is the greater, for $D/\lambda \geq 50$.

$\varphi_{min}$ = 2° or 114 $(D/\lambda)^{-1,09}$ degrees, whichever is the greater, for $D/\lambda < 50$.

D is the nominal diameter of the antenna.

In addition the maximum antenna gain of each of the cross-polarized components in any direction ϕ degrees from the antenna main beam axis shall not exceed the following limits:

$$G_x(\varphi) = 23 - 20 \log \varphi \quad \text{dBi} \quad \text{for } \varphi_r \leq \varphi \leq 7°$$

where $\varphi_r$ is equal to 1° or 100 λ/*D*, whichever is greater.

If the receive antenna performance does not meet the above requirement, then the earth station shall accept an additional co frequency interference caused by its off-axis gain which is not compliant to the mask.

#### 4.2.9.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.12.

### 4.2.10 Blocking performance

#### 4.2.10.1 Justification

To prevent high power signals outside the receive frequency band from blocking the reception of signals inside the receive frequency band.

#### 4.2.10.2 Specification

Receiver blocking is characterized here through gain compression for a signal inside the receive frequency band that is caused by another signal outside the receive frequency band at high power. The level of the other signal is compared to the level of a signal inside the receive frequency band that would cause the same gain compression.

Receiver blocking rejection at a particular frequency is defined as the level of a second signal at this frequency that causes a certain gain compression to a first signal inside the receive frequency band, minus the level of a second signal at a frequency inside the receive frequency band that causes the same gain compression.

The first signal shall be the wanted modulated signal as defined for the intended use, tuned at the centre frequency of the receive frequency band. The second signal shall be a CW signal with a level of 30 dB above the reference sensitivity level and shall not cause a gain compression for the first signal higher than 1 dB.

The rejection shall comply with table 4.

**Table 4: Receiver blocking rejection**

| Frequency | Minimum rejection |
|---|---|
| below 9 GHz | 20 dB |
| 9 to 10 GHz | 10 dB |
| 14 to 16 GHz | 10 dB |
| Above 16 GHz | 20 dB |

#### 4.2.10.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.13.

### 4.2.11 Adjacent Signal Selectivity

#### 4.2.11.1 Justification

To enable reception of a wanted signal in presence of other signals on adjacent frequencies which are transmitted with high EIRP density from target satellite orbital position.

NOTE: The power level of signals transmitted from the same orbital position are under control of the satellite operator. Signals transmitted from an adjacent orbital position that is not near-by are suppressed by the antenna gain pattern.

#### 4.2.11.2 Specification

Adjacent Signal Selectivity is a measure of a receiver's ability to receive a signal at its assigned channel frequency in the presence of an adjacent signal at a given frequency offset from the centre frequency of the assigned channel.
The adjacent signal shall occupy the same bandwidth as the wanted signal. Frequency offset and relative power level of the adjacent signal compared to the wanted signal shall take the value given in table 5. BW is the wanted signal occupied bandwidth.

**Table 5: Adjacent signal frequency and Power level**

| Signal | Centre frequency offset from wanted signal | Power level relative to wanted signal |
|---|---|---|
| Adjacent Carrier | BW | 7 dBsd |

The decrease in the required signal to noise ratio in the presence of an adjacent signal shall be no more than 0,5 dB.

#### 4.2.11.3 Conformance tests

Conformance tests shall be carried out in accordance with clause 6.14.

# 5 Testing for compliance with technical requirements

## 5.1 Environmental conditions for testing

Tests defined in the present document shall be carried out at representative points within the boundary limits of the operational environmental profile defined by its intended use.

Where technical performance varies subject to environmental conditions, tests shall be carried out under a sufficient variety of environmental conditions (within the boundary limits of the operational environmental profile defined by its intended use) to give confidence of compliance for the affected technical requirements.

# 6 Test methods for all aspects of the EST

## 6.1 General

The present clause describes the general requirements for verifying that compliance of the EUT against the requirements. The documentary evidence of the results obtained in performing the essential radio tests shall be maintained with the documentation of the EUT. At a minimum, the documentary evidence shall include:

a) test setup (configurations, test equipment and calibration status);

b) test conditions (environmental and operational parameters);

c) method of testing;

d) results of measurements, measurement resolution & uncertainty; and

e) statement of compliance with technical specifications of the present document.

The test conditions for each test shall be representative of the intended operational environment and be noted in the test report where it has a significant effect on the measurement results.

The type, termination and length of each cable used shall be representative of the intended installation and recorded in the test report. For maximum cable lengths longer than 10 m, as defined for the intended use of the EST, the tests shall be performed with cables no shorter than 10 m.

ETSI TR 102 215 [i.4] provides a recommended approach to measurement uncertainty and interpretation of results. Where the measurement approach does not conform to the guidance in ETSI TR 102 215 [i.4], the EUT shall be provided with documentation to justify the uncertainty achieved under the measurement conditions.

The test site and the method of measurement shall also be in accordance with CISPR 16-1-5 [3] as applicable.

To enable the tests to be carried out the use of an NCF or a Special Test Equipment (STE), made available with the EUT, may be necessary. Since this STE will be specific for the particular system, it is not possible to provide detailed specifications in the present document. However, the following baseline is provided:

a) if the EST requires to receive a modulated carrier from the satellite in order to transmit, then special test arrangements are required to simulate the satellite signal, thus enabling the EST to transmit allowing measurement of transmission parameters;

b) any characteristic of these special test arrangements which may have direct or indirect effects on the parameters to be measured shall be clearly defined for the intended use of the EST.

All tests with carrier-on shall be undertaken with the transmitter operating at $EIRP_{max}$, as per the specific requirement, and with the normal radio operating parameters, as defined for the intended use of the EST.

If the Equipment Under Test (EUT) is an EST that requires hardware and/or software modification(s) for these tests then full documentation of such modification(s) shall be provided to demonstrate that the modification(s) will simulate the required test condition, without its main characteristics being changed.

# 6.2 Off-axis spurious radiation

## 6.2.0 General

The tests for the EST specification in clause 4.2.1.2 shall be conducted in "Carrier on", "Carrier off" and "Emissions disabled" radio states as required. The tests in "Carrier on" radio state shall be undertaken with the transmitter operating at $EIRP_{max}$.

## 6.2.1 Test method

An EUT with antenna is an EST with its antenna. It comprises both the internal and external units interconnected by the cable(s). It comprises both the internal and external units, up to the antenna flange, interconnected by the cable(s). The connecting cable(s) between the internal and external units shall be the same types as defined for the intended use of the EST. The type of each cable used shall be entered in the test report.

For frequencies up to 80 MHz the measuring antenna shall be a balanced dipole with a length equal to the 80 MHz resonant length and shall be matched to the feeder by a suitable transforming device. Measurements with broad band antennas are also possible provided that the test site has been calibrated according to the requirements of CISPR 16-1-5 [3].

For frequencies between 80 MHz and 1 000 MHz the measuring antenna shall be a balanced dipole which shall be resonant in length. Measurements with broad band antennas are also possible provided that the test site has been calibrated according to the requirements of CISPR 16-1-5 [3].

For frequencies above 1 000 MHz the antenna shall be a horn radiator of known gain/frequency characteristics. When used for reception the antenna and any associated amplification system, the combined amplitude/frequency response calibration curves shall be verified across the measurement frequency range considered for the antenna, to account for the interaction between the components. The antenna is mounted on a support capable of allowing the antenna to be used in either horizontal or vertical polarization and at the specified height.

Unused input and output ports of the EUT shall be connected to cables (see clause 6.1) and terminated to simulate the impedance of the relevant ports of the radio equipment. These cables shall be bundled at the approximate centre of the cable with the bundles of 30 cm to 40 cm in length running in the horizontal plane from the port to which it is connected. If it is impractical to do so because of cable bulk or stiffness, the disposition of the excess cable shall be precisely noted in the test report.

The radiations shall be measured in the frequency range of 30 MHz to 2 GHz in accordance with CISPR 16-1-5 [3] using the measuring receiver or a comparable spectrum analyser. During the measurements up to 1 GHz the quasi-peak detector shall be used, for measurements in the frequency range 1 GHz to 2 GHz a peak detector shall be used.

# 6.3 On-axis spurious radiation

## 6.3.0 General

The tests for the EST specification in clause 4.2.2.2 shall be conducted in "Carrier on", "Carrier off" and "Emissions disabled" radio states as required.

The tests in "Carrier on" radio state shall be undertaken with the transmitter operating at $EIRP_{max}$. Test measurements shall be made at least three points within the operating band including: locating the carrier at the centre frequency of the operating band; as close to the lower operating frequency limit as possible; and as close to the upper operating frequency limit as possible.

## 6.3.1 Test method

### 6.3.1.1 General

The tests shall be undertaken with the transmitter operating at $EIRP_{max}$.

For EST equipment for which measurements at the antenna flange are not possible, the measurements shall be performed with a test antenna.

For EST equipment for which measurements at the antenna flange are possible, the measurements shall be performed at the antenna flange. The EUT is the EST with its antenna comprising both the internal and external units interconnected by appropriate cables.

### 6.3.1.2 Method of measurement at the antenna flange

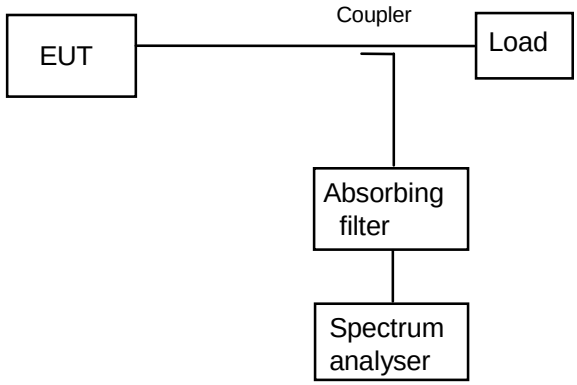



**Figure 3: Test set up - on-axis spurious radiation measurements at the antenna flange**

a) The test set up shall be as shown in figure 3. In order to protect the spectrum analyser an absorbing filter can be used and it shall be tuned to the transmit carrier frequency.

b) The EUT shall transmit one modulated carrier continuously at its maximum burst rate, centred on a frequency within the lower limit of the operating frequency band of the EUT. The EUT shall be operated at the highest normal operating e.i.r.p. The frequency range 14,00 GHz to 14,25 GHz shall be measured.

c) Due to the proximity of the carrier the spectrum analyser resolution bandwidth shall be set to a measurement bandwidth of 3 kHz, or as close as possible. If the measurement bandwidth is different from the specified measurement bandwidth, bandwidth correction shall be performed for noise-like wideband spurious radiation.

d) To obtain the on-axis spurious e.i.r.p., the antenna transmit gain shall be added to any figure obtained in the above measurement and any correction or calibration factor summated with the result. The antenna gain shall be as measured in clause 6.3.1.3 at the closest frequency to the spurious frequency.

e) The tests in b) to e) shall be repeated with a transmit frequency in the centre of the operating frequency band.

f) The tests in b) to e) shall be repeated with a transmit frequency as close to the upper limit of the operating frequency band of the EUT as possible.

g) The test shall be repeated in the "Carrier-off" radio state.

h) The test shall be repeated in the "Emissions disabled" radio state.

#### 6.3.1.3 Method of measurement with a test antenna

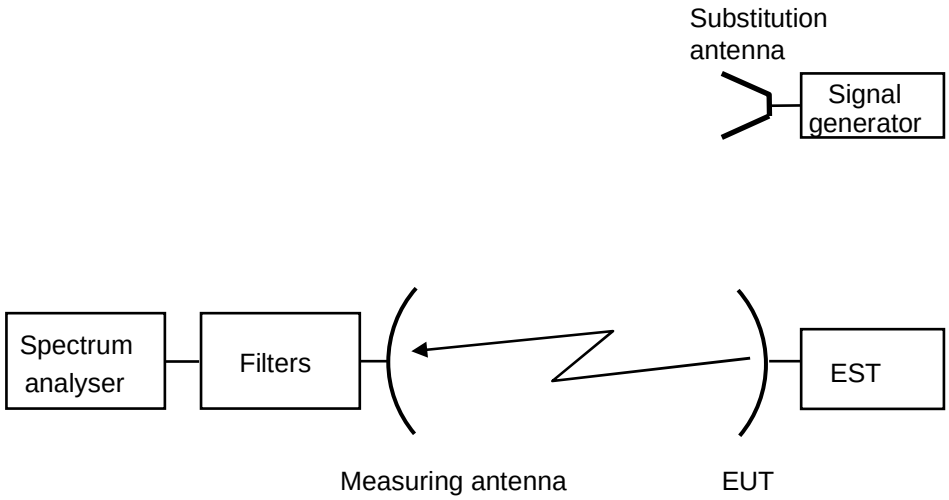



**Figure 4: Test set up - on-axis spurious radiation measurements for an EUT with antenna**

a) The test set up shall be as shown in figure 4.

b) The EUT shall be installed such that the units are separated by about 1 m to 2 m with the internal unit at a height between 0,5 m and 1,0 m on a turntable. The interconnection cable shall be maintained by non-metallic means at a height between 0,5 m and 1,0 m.

c) The spectrum analyser resolution bandwidth shall be set to the specified measuring bandwidth or as close as possible. If the resolution bandwidth is different from the specified measuring bandwidth, bandwidth correction shall be performed for noise-like wideband spurious radiation.

d) The EUT shall transmit the modulated carrier(s) continuously, or at its maximum burst rate where applicable, centred on a frequency as close to the lower limit of the operating frequency band of the EUT as possible. The EUT shall be operated at $EIRP_{max}$. The frequency range 14,00 GHz to 14,25 GHz shall be measured and each spurious frequency shall be noted.

e) Due to the proximity of the carrier the spectrum analyser resolution bandwidth shall be set to a measurement bandwidth of 3 kHz, or as close as possible. If the measurement bandwidth is different from the specified measurement bandwidth, linear bandwidth correction shall be performed for noise-like wideband spurious radiation.

f) The measuring antenna shall be positioned at a distance from the EUT (e.g. 3 m, 5 m, 10 m) relevant to the applied test site and shall be aligned with the EUT antenna for the transmit frequency. The measuring antenna shall be adjusted in height, whilst the EUT is in the appropriate carrier condition, for a maximum response on the associated spectrum analyser at each spurious frequency previously identified, this response level shall be noted. The adjustment in height of the measuring antenna does not apply when an anechoic chamber is being used.

g) The EUT shall be replaced by a representative substitution antenna to which is connected a signal generator. The main beam axes of the measuring and substitution antennas shall be aligned. The distance between these antennas shall be the distance determined under test f).

h) The substitution and measuring antennas shall be aligned in the polarization which produced the largest response between the EUT and the test antenna.

i) The output of the generator shall be adjusted so that the received level is identical to that of the previously noted largest spurious radiation.

j) The output level of the signal generator shall be noted. The e.i.r.p. of the on-axis spurious radiation is the sum, in dB, of the signal generator output plus the substitution antenna isotropic gain minus the interconnection cable loss.

k) The tests in d) to j) shall be repeated with a transmit frequency in the centre of the operating frequency band.

l) The tests in d) to j) shall be repeated with a transmit frequency as close to the upper limit of the operating frequency band of the EUT as possible.

m) The test shall be repeated in the "Carrier-off" radio state.

n) The test shall be repeated in the "Emissions disabled" radio state.

# 6.4 Off-axis e.i.r.p. emission density within the band

## 6.4.0 General

The tests for the EST specification in clause 4.2.3.2 for off-axis e.i.r.p. emission density (co-polarization and cross-polarization) within the band 14,0 GHz to 14,25 GHz shall be conducted in the "Carrier on" radio state. The tests shall be undertaken with the transmitter operating at $EIRP_{max}$.

The tests in "Carrier on" radio state shall be undertaken with the transmitter operating at $EIRP_{max}$. Test measurements shall be made at least three points within the operating band including: locating the carrier at the centre frequency of the operating band; as close to the lower operating frequency limit as possible; and as close to the upper operating frequency limit as possible.

## 6.4.1 Test method

### 6.4.1.1 General

The measurement shall be made in the operational configuration(s) (including occupied bandwidth, $EIRP_{max}$) producing the highest emission density in the frequency band. The selected configuration(s) shall be recorded in the test report.

Test with an STE can be done; then the STE shall provide all the signals needed by the ESTs under normal operation (e.g. a radio beacon if it is expected to be received by the satellite).

The e.i.r.p. densities are determined from the measurements of the antenna co-polarization and cross-polarization gain patterns, and of the power density at the antenna flange. These e.i.r.p. densities shall be compared to the specified masks of clause 4.2.3.2.

To ascertain the off-axis e.i.r.p. it is necessary to know the transmit power density and antenna transmit radiation pattern. To ascertain the radiation pattern it is necessary to know the antenna transmit gain.

The following three measurement procedures shall, therefore, be performed:

a) transmit output power density (dBW/40 kHz);

b) antenna transmit gain (dBi);

c) antenna transmit radiation patterns (dBi).

### 6.4.1.2 Transmit output power density

#### 6.4.1.2.1 General

The EUT output power shall be set to the power corresponding to $EIRP_{max}$.

For the purpose of this test the EUT is defined as the internal unit and that part of the external unit up to the antenna flange.

#### 6.4.1.2.2 Test site

There are no requirements for the test site to be used for this test.

#### 6.4.1.2.3 Method of measurement

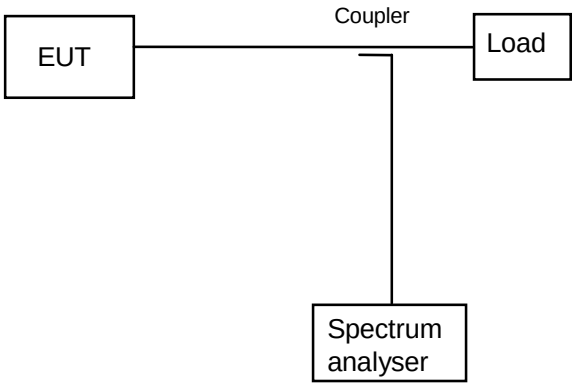



**Figure 5: Test set up - transmit output power density measurement**

a) The test set up shall be as shown in figure 5.

b) The EUT shall transmit the carrier(s) with a pseudo random signal. For burst mode transmission the EUT shall transmit at its maximum burst-rate. The power density supplied to the antenna flange shall be measured in dBW/40 kHz. The coupling factor of the test coupler at the test frequency and the attenuation of any necessary waveguide adapter shall be taken into account.

   The spectrum analyser shall be operated under the following conditions:

   a) Frequency span:
      As required for the complete nominated bandwidth.

   b) Resolution bandwidth:
      The resolution bandwidth of the spectrum analyser shall be set as close as possible to the specified measuring bandwidth of 40 kHz. If the resolution bandwidth is different from the specified bandwidth then bandwidth correction shall be performed except for spectrum components the width of which are narrower than the measurement bandwidth.

   c) Display/video bandwidth:
      Equal to the resolution bandwidth.

   d) Averaging:
      Yes.

   e) Max hold:
      No.

   If so provided with the EUT, the test may be performed in max hold mode.

   For an EST operating in a continuous carrier mode, the measurement time shall be sufficient to ensure that at any frequency the difference between any two measurement results is less than 1 dB.

   For an EST operating in a non-continuous carrier mode, the average measurement shall be performed over the transmitted burst and the measurements from a series of bursts shall be combined as follows:

   a) Each measurement can exclude a specific portion of each burst. The excluded portion shall not exceed 50 µs or 10 % of the burst, whichever is the smaller. The excluded portion shall be defined for the intended use of the EST.

b) The measurements from several bursts shall be averaged to give a final measurement result. The number of bursts that are averaged shall be sufficient to ensure that the difference between any two final measurement results is less than 1 dB.

c) The measurement shall be made in the operational configuration(s) (occupied bandwidth, $EIRP_{nom}$) producing the highest emission density in the frequency band. The selected configuration(s) shall be recorded in the test report.

### 6.4.1.3 Antenna transmit gain

#### 6.4.1.3.1 General

For the purpose of the present document, the antenna transmit gain is defined as the ratio, expressed in decibels (dBi), of the power that would have to be supplied to an isotropic radiator isolated in space to the power supplied to the antenna being considered, so that they produce the same field strength at the same distance in the same direction. Unless otherwise specified the transmit gain is for the direction of maximum radiation (i.e. peak transmit gain).

For the purpose of this test the EUT is defined as that part of the external unit which comprises the antenna and its flange. The antenna includes the reflector(s), feed, support struts and an enclosure of equal weight/distribution to any electrical equipment normally housed with the feed at the antenna focal point, as the EST is planned to be brought to the market.

#### 6.4.1.3.2 Test site

This test shall be performed using the following method of measurement and test set-up as specified in the following clauses.

The tests can be carried out in an Open Area Test Site (OATS) environment or in an anechoic chamber or in a semi-anechoic chamber as they lead to comparable test results for establishing compliance. Ambient noise levels shall be at least 6 dB below the applicable unwanted emissions limit. The preferred test site is the semi-anechoic chamber.

#### 6.4.1.3.3 Method of measurement

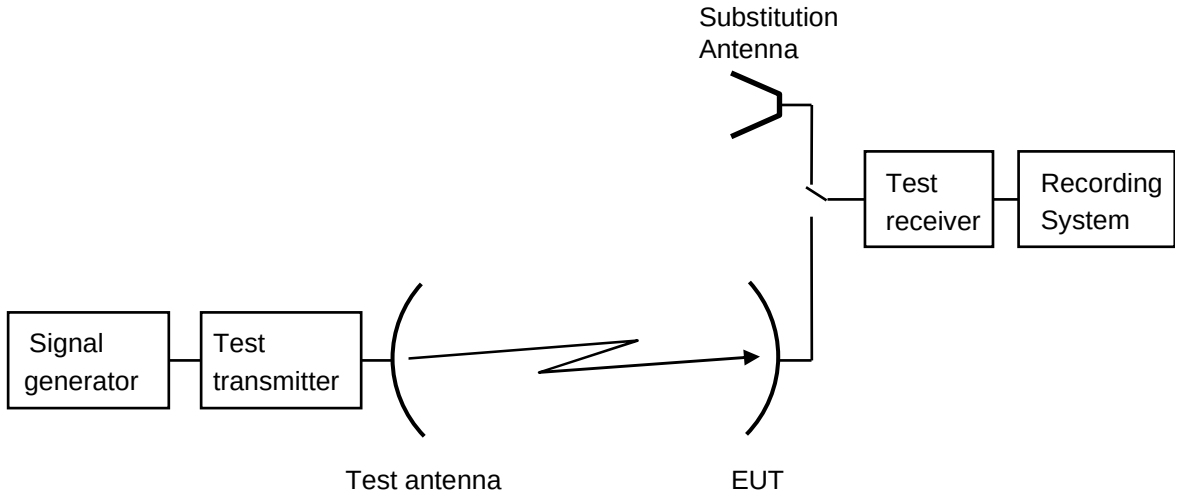



**Figure 6: Test set up - antenna transmit gain measurement**

a) The test set up shall be as shown in figure 6 with the EUT connected to the test receiver. The test arrangement shall enable the test results to be displayed and recorded with the angular position and the signal level.

b) A test signal shall be transmitted by the test transmitter through the test antenna. The centre frequency of the test signal shall be set to the middle of the operating frequency band as defined for the intended use of the EST The E-plane shall be vertical. The EUT antenna main beam axis shall be aligned with the main beam axis of the test transmitter. The polarizer of the EUT antenna shall be rotated and adjusted such that the E-plane coincides with the E-plane of the test transmitter.

c) The EUT shall be aligned to maximize the received signal and the recording system adjusted to give the maximum reading on the plot.

d) The EUT shall be driven in azimuth in one direction through 10°.

e) The pattern measurement is then obtained by driving the EUT in azimuth back through boresight to 10° the other side and recording the results.

f) The EUT shall be replaced by the substitution antenna and the received signal level maximized.

g) This level shall be recorded.

h) The substitution antenna shall be driven in azimuth as in d) and e).

i) The gain of the EUT shall be calculated from:

$$G_{EUT} = L_1 - L_2 + C$$

where:

- $G_{EUT}$ is the gain of the EUT (dBi);
- $L_1$ is the level obtained with the EUT (dB);
- $L_2$ is the level obtained with the substitution antenna (dB);
- C is the calibrated gain of the substituted antenna at the test frequency (dBi).

j) The tests in c) to i) shall be repeated with the frequency changed to lowest carrier centre frequency of the operating frequency band as defined for the intended use of the EST.

k) The tests in c) to i) shall be repeated with the frequency changed to highest carrier centre frequency of the operating frequency band as defined for the intended use of the EST.

### 6.4.1.4 Antenna transmit radiation patterns

#### 6.4.1.4.1 General

For the purpose of the present document, the antenna transmit radiation patterns are diagrams relating field strength to the angle of the direction pointed by the antenna at a constant large distance from the antenna (in the far field of the antenna).

#### 6.4.1.4.2 Test site

This test shall be performed using the following method of measurement and test set-up as specified in the following clauses.

The tests can be carried out in an Open Area Test Site (OATS) environment or in an anechoic chamber or in a semi-anechoic chamber as they lead to comparable test results for establishing compliance. Ambient noise levels shall be at least 6 dB below the applicable unwanted emissions limit. The preferred test site is the semi-anechoic chamber.

#### 6.4.1.4.3 Test arrangement

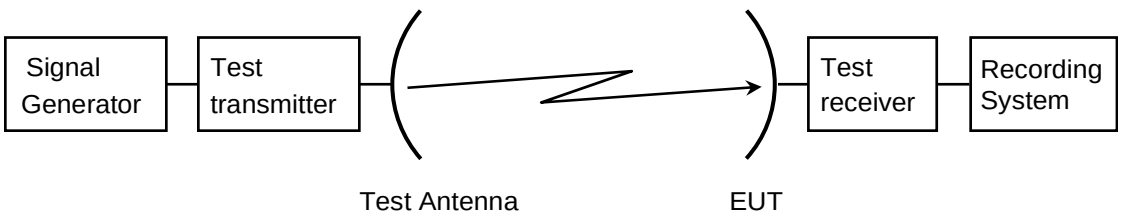



**Figure 7: Test set up - antenna transmit radiation pattern measurement**

#### 6.4.1.4.4 Co-polar radiation pattern-azimuth

a) The test set up shall be as shown in figure 7 with the EUT connected to the test receiver. The test set up shall enable the test results to be displayed and recorded with the angular position along and the signal level.

b) The frequency of the test signal shall be set to the middle frequency of the operating frequency band as defined for the intended use of the EST.

c) The initial E-plane of the test signal radiated by the test transmitter through its antenna shall be vertical. The EUT antenna main beam axis shall be aligned with the main beam axis of the test transmitter. The polarizer of the EUT antenna shall be rotated and adjusted such that its E-plane coincides with the E-plane of the test transmitter. Precise co-polar peaking of the polarization shall be done by observing the cross-polar minimum (fine adjustment).

d) The EUT shall be aligned to maximize the received signal and the recording system adjusted to give the maximum reading on the plot.

e) The EUT shall be driven in azimuth to -180°.

f) The transmit pattern measurement is then obtained by driving the EUT in azimuth from -180° to +180° with the results recorded.

g) The tests in d) to f) shall be repeated with the frequency changed to the lowest carrier centre frequency of the operating frequency band as defined for the intended use of the EST.

h) The tests in d) to f) shall be repeated with the frequency changed to the highest carrier centre frequency of the operating frequency band as defined for the intended use of the EST use.

i) The tests in b) to h) may be performed simultaneously.

j) The tests in d) to i) shall be repeated with the E-plane of the test signal being horizontal. The frequency of the test signal shall be set to middle of the operating frequency band as defined for the intended use of the EST. The polarizer of the EUT antenna shall be rotated and adjusted such that its E-plane coincides with the E-plane of the test transmitter (giving the minimum cross-polar receive signal). Precise co-polar peaking of the polarization shall be done by observing the cross-polar minimum.

#### 6.4.1.4.5 Co-polar radiation pattern-elevation

a) The test set up shall be as shown in figure 7 with the EUT connected to the test receiver. The test arrangement shall enable the test results to be displayed and recorded with the angular position and the signal level.

b) The frequency of the test signal shall be set to the middle frequency of the operating frequency band as defined for the intended use of the EST.

c) The initial E-plane of the test signal radiated by the test transmitter through its antenna shall be vertical. The EUT antenna main beam axis shall be aligned with the main beam axis of the test transmitter. The polarizer of the EUT antenna shall be rotated and adjusted such that its E-plane coincides with the E-plane of the test transmitter. Precise co-polar peaking of the polarization shall be done by observing the cross-polar minimum (fine adjustment).

d) The EUT shall be aligned to maximize the received signal and the recording system adjusted to give the maximum reading on the plot.

e) The EUT shall be driven in elevation to -1°.

f) The transmit pattern measurement is then obtained by driving the EUT in elevation from -1° to 70° and the results recorded.

g) The tests in d) to f) shall be repeated with the frequency changed to the lowest carrier centre frequency of the operating frequency band as defined for the intended use of the EST.

h) The tests in d) to f) shall be repeated with the frequency changed to the highest carrier centre frequency of the operating frequency band as defined for the intended use of the EST.

i) The tests in b) to h) may be performed simultaneously.

j) The tests in d) to i) shall be repeated with the E-plane of the test signal being horizontal. The frequency of the test signal shall be set to the middle frequency of the operating frequency band as defined for the intended use of the EST. The polarizer of the EUT antenna shall be rotated and adjusted such that its E-plane coincides with the E-plane of the test transmitter. The EUT antenna main beam axis shall be aligned with the main beam axis of the test transmitter. Precise co-polar peaking of the polarization plane shall be done by observing the cross-polar minimum (fine adjustment).

#### 6.4.1.4.6 Cross-polar radiation pattern-azimuth

a) The test set up shall be as shown in figure 7 with the EUT connected to the test receiver. The test set up shall enable the test results to be displayed and recorded with the angular position and the signal level.

b) The frequency of the test signal shall be set to the middle frequency of the operating frequency band as defined for the intended use of the EST.

c) The initial E-plane of the test signal radiated by the test transmitter through its antenna shall be vertical. The EUT antenna main beam axis shall be aligned with the main beam axis of the test transmitter. The polarizer of the EUT antenna shall be rotated and adjusted such that its E-plane is orthogonal to the E-plane of the test transmitter. Precise fine adjustment of the polarization plane shall be done by observing the cross-polar minimum.

d) The boresight co-polar receive signal shall be used to adjust the recording system such that it gives the maximum reading on the plot.

e) The EUT shall be driven in azimuth to -10°.

f) The transmit pattern measurement is then obtained by driving the EUT in azimuth from -10° to +10° with the results recorded.

g) The tests in d) to f) shall be repeated with the frequency changed to the lowest carrier centre frequency of the operating frequency band as defined for the intended use of the EST.

h) The tests in d) to f) shall be repeated with the frequency changed to the highest carrier centre frequency of the operating frequency band as defined for the intended use of the EST.

i) The tests in b) to h) may be performed simultaneously.

j) The tests in d) to i) shall be repeated with the E-plane of the test signal being horizontal. The frequency of the test signal shall be set to the middle frequency of the operating frequency band as defined for the intended use of the EST. The EUT antenna main beam axis shall be aligned with the main beam axis of the test transmitter. The polarizer of the EUT antenna shall be rotated and adjusted such that its E-plane is orthogonal with the E-plane of the test transmitter precise fine adjustment of the polarization plane shall be done by observing the cross-polar minimum.

#### 6.4.1.4.7 Cross-polar radiation pattern-elevation

a) The test set up shall be as shown in figure 7 with the EUT connected to the test receiver. The test set up shall enable the test results to be displayed and recorded with the angular position and the signal level.

b) The frequency of the test signal shall be set to the middle frequency of the operating frequency band as defined for the intended use of the EST.

c) The initial E-plane of the test signal radiated by the test transmitter through its antenna shall be vertical. The EUT antenna main beam axis shall be aligned with the main beam axis of the test transmitter. The polarizer of the EUT antenna shall be rotated and adjusted such that its E-plane is orthogonal to the E-plane of the test transmitter. Precise fine adjustment of the polarization plane shall be done by observing the cross-polar minimum.

d) The boresight co-polar receive signal shall be used to adjust the recording system such that it gives the maximum reading on the plot.

e) The EUT shall be driven in elevation to -1°.

f) The transmit pattern measurement is then obtained by driving the EUT in elevation from -1° to +10° with the results recorded.

g) The tests in d) to f) shall be repeated with the frequency changed to the lowest carrier centre frequency of the operating frequency band as defined for the intended use of the EST.

h) The tests in d) to f) shall be repeated with the frequency changed to the highest carrier centre frequency of the operating frequency band as defined for the intended use of the EST.

i) The tests in b) to h) may be performed simultaneously.

j) The tests in d) to i) shall be repeated with the E-plane of the test signal being horizontal. The frequency of the test signal shall be set to the middle frequency of the operating frequency band as defined for the intended use of the EST. The EUT antenna main beam axis shall be aligned with the main beam axis of the test transmitter. The polarizer of the EUT antenna shall be rotated and adjusted such that its E-plane is orthogonal with the E-plane of the test transmitter precise fine adjustment of the polarization plane shall be done by observing the cross-polar minimum.

### 6.4.2 Computation of results

The results shall be computed by producing a "mask" to the specified limits with the reference level being equal to the sum of the transmitter output power density and the gain of the antenna. This reference shall then be placed on the maximum point of the plots obtained from the transmit radiation pattern measurements, so as to ascertain that the off-axis e.i.r.p. density is within the mask, and thus conforming to the specification in clause 4.2.3.

## 6.5 Carrier suppression

### 6.5.1 Test method

For the purposes of this test, the EUT shall transmit one carrier modulated continuously, or at its maximum burst rate, centred on the middle frequency of the operating frequency band as defined for the intended use of the EST. The "transmission disabled state" shall be obtained by use of the NCF or an STE. For conducted measurements the maximum residual carrier power density within the nominated bandwidth shall be measured and added to the antenna on-axis gain. For radiated measurements the maximum residual e.i.r.p. density within the nominated bandwidth shall be measured and recorded:

a) The test set up for radiated measurements shall be as shown in figure 7.

b) The EUT shall transmit one modulated carrier continuously, or at its maximum burst rate, centred on the middle frequency of the operating frequency band as defined for the intended use of the EST.

c) The resolution bandwidth of the spectrum analyser shall be set to 3 kHz.

d) The "transmission disabled state" shall be obtained by use of the NCF.

e) For conducted measurements the maximum residual carrier power density within the nominated bandwidth shall be measured and added to the antenna on-axis gain.

f) For radiated measurements the maximum residual e.i.r.p. density within the nominated bandwidth shall be measured and recorded.

An STE provided with the EUT can be used to enable the EST transmission to be suppressed.

## 6.6 Antenna pointing

### 6.6.1 General

The antenna pointing test method is designed to verify the ability of the EUT to detect and correctly respond to pointing errors that exceed the threshold, $\delta\phi$, defined for the intended use of the EST as required in clauses 4.2.3 and 4.2.5.

For the purpose of these tests the EUT is the EST with its antenna connected.

## 6.6.2 Test method

The means for inducing pointing errors shall be representative of actual operating train dynamics, within the dynamic limits defined for the intended use of the EST.. Pointing errors shall be induced in at least two orthogonal directions; but not necessarily during the same test sequence. The means of inducing pointing errors shall be included in the test report.

After the pointing error has been decreased below the pointing error threshold, it shall be verified that the EUT does not re-enter the "Carrier-on" radio state for at least $2 \times T$ seconds, as specified in clause 4.2.5.

The tests shall be repeated 5 times to ensure reliability of pointing error detection and appropriate response.

The antenna pointing test method is designed to verify the ability of the EUT to detect and correctly respond to pointing errors that exceed the threshold, $\delta\phi$, defined for the intended use of the EST as required in clause 4.2.5. For the purposes of testing antenna pointing, the documentation provided with the EUT shall include:

a) the operational dynamic limits for ship motion under which the EST can maintain tracking within the defined $\delta\phi$;

b) the survivability dynamic limits for operation without permanent damage; and

c) the dynamic limits for testing of the EUT, which shall not be less than the operational dynamic limits and not greater than the survivability dynamic limits.

For the purpose of these tests the EUT is the EST with its antenna connected. The test set up is provided in the following figure 8.

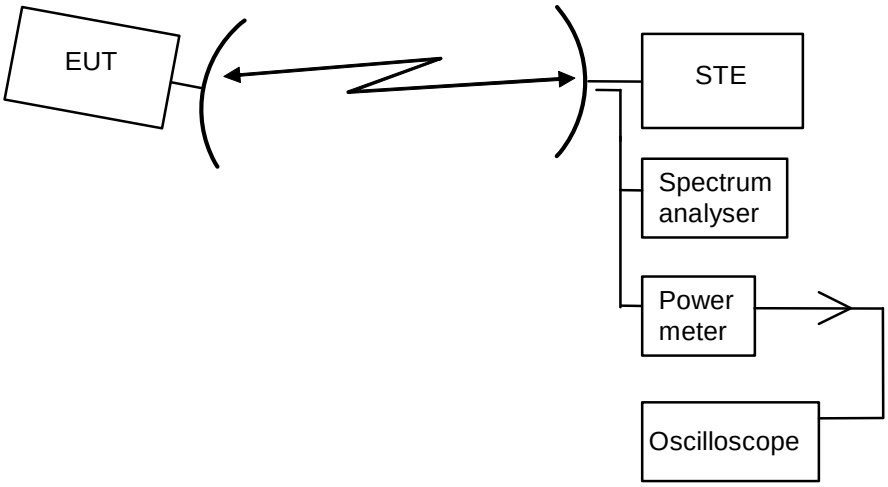



**Figure 8: Test set up for pointing error detection**

a) The equipment shall be tested as shown in figure 8. An actual satellite link may be used in place of the STE. In the case of using an actual satellite link, initial step shall be taken so that interference to adjacent satellites does not occur due to the pointing errors that will be induced in the test procedure (e.g. suppress the carrier or use a low e.i.r.p.).

b) The signal received by the EUT shall be adjusted so that the signal to noise ratio corresponds to the minimum threshold signal to noise ratio defined for the intended use of the EST for reliable tracking.

c) The spectrum analyser or the power meter shall be used to verify that the EUT has suppressed the carrier.

d) The true pointing error values defined for the intended use of the EST shall be measured and analysed.

e) The EUT shall be capable to dynamically induce pointing errors that exceed the threshold defined for the intended use of the EST. The induced pointing errors shall be representative of operating train dynamics, within the dynamic limits for testing the EUT, as defined for the intended use of the EST in clause 6.7.1. Pointing errors shall be induced in at least two orthogonal directions; but not necessarily during the same test sequence.

f) Conduct of test:

1) EUT shall be switched on and pointing towards the STE in a stable tracking state.

2) Pointing errors of an increasing magnitude shall be induced in the EUT using the test set up e) above until the defined pointing error threshold is exceeded.

3) It shall be verified that the EUT correctly detects that the induced pointing error has exceeded the pointing error threshold and enters the "Carrier-off" radio state no later than T seconds after the pointing error threshold is exceeded, where T is the response time defined for the intended use of the EST.

4) Using the dynamic means given in e), the induced pointing error shall be decreased until it is less than the pointing error threshold.

5) It shall be verified that the EUT does not re-enter the "Carrier-on" radio state for at least $2 \times T$ seconds.

g) Step f) shall be repeated ten times.

h) If the results in f) are satisfied on all ten tests, then the EUT shall be declared compliant with the requirements

# 6.7 Antenna mechanical stability

## 6.7.1 Test method

The methodology described in annex B shall be used to show compliance with the specification for mechanical stability.

# 6.8 Void

# 6.9 Cessation of emissions of the EST

## 6.9.0 General

There shall be a means of simulating a condition where cessation of emissions is required. The spectrum analyser or the oscilloscope may be used to measure the time difference between the occurrence of the condition and the cessation of emissions. In all test methods, it shall be verified that the time difference does not exceed the value as specified in clause 4.2.6.2.3.

Where the EUT adds an operational margin around areas where cessation of emissions is required to account for the position determination accuracy and latency, the test procedure may simulate the actual behaviour of an EUT as it enters into an area where cessation of emissions is required. For example, if the EUT adds a margin of X metres around areas where cessation of emissions is required, then the test may begin with the EUT being simulated at a position more than X metres outside of an area where cessation of emissions is required. The test may then simulate the motion of the EUT towards the area where cessation of emissions is required. In any event, the EUT shall enter the "emission disabled" radio state of entering the actual area where cessation of emissions is required, as specified in clause 4.2.6.2.3.

For the purpose of these tests the EUT is the EST either with or without its antenna connected.

## 6.9.1 Test Method

### 6.9.1.1 Required documentation

The EUT documentation shall provide the evidence to verify the mode(s) of cessation of emissions implemented in the EUT and the EST interfaces involved in the cessation of emissions as specified in clause 4.2.6.2.1.

The EUT documentation shall provide the evidence to verify the compliance with specification 2 in clause 4.2.6.2.2 for the determination of the conditions under which the EST shall cease emissions.

#### 6.9.1.2 Cessation of emissions from the "Transmission enabled" state

a) The EUT shall be set in the "Transmission enabled" EST state.

b) The EUT shall be set in the "Carrier on" radio state.

c) A condition requiring the cessation of emission shall be initiated either by the STE or the EUT.

d) It shall be verified that the EUT enters the "Emissions disabled" radio state.

e) In order to verify that the EUT is in the "Transmission disabled" EST state, it shall be verified that the EUT remains in the "Emissions disabled" radio state when the EUT is requested to transmit data.

#### 6.9.1.3 Cessation of emission from the "Transmission disabled" state

a) The EUT shall be set in the "Transmission disabled" state.

b) A condition requiring the cessation of emission shall be initiated either by the STE or the EUT.

c) In order to verify that the EUT remains in the "Transmission disabled" EST state, it shall be verified that the EUT remains in the "Emissions disabled" radio state when the EUT is requested to transmit data.

d) The STE shall send a TxE command to the EUT.

e) In order to verify that the EUT remains in the "Transmission disabled" EST state, it shall be verified that the EUT remains in the "Emissions disabled" radio state when the EUT is requested to transmit data.

f) The STE shall send a TxD command to the EUT.

g) The condition requiring the cessation of emissions shall be removed.

h) In order to verify that the EUT remains in the "Transmission disabled" EST state, it shall be verified that the EUT remains in the "Emissions disabled" radio state when the EUT is requested to transmit data.

#### 6.9.1.4 Cessation of emission from the "Initial Phase" state

##### 6.9.1.4.1 EUTs transmitting initial bursts

This applies only to EUTs transmitting initial bursts (i.e. where "Initial phase-BurstOn" and "Initial phase-BurstOff" are implemented):

a) The EUT shall be set in "Initial phase-BurstOn" state.

b) The STE shall not send the TxE command and the EUT shall be allowed to cycle between "Initial phase-BurstOn" and "Initial phase-BurstOff" as it would under normal operations.

c) A condition requiring the cessation of emission shall be initiated either by the STE or the EUT.

d) It shall be verified that the EUT enters the "Emissions disabled" radio state.

e) In order to verify that the EUT is in the "Initial phase-Standby" EST state, it shall be verified that the EUT remains in the "Emissions disabled" radio state when the EUT is requested to transmit data.

f) The STE shall send a TxE command to the EUT.

g) In order to verify that the EUT remains in the "Initial phase-Standby" EST state or enters the "Transmission disabled" EST state, it shall be verified that the EUT remains in the "Emissions disabled" radio state when the EUT is requested to transmit data.

h) Repeat the test steps a) through g) using a TxD command in step f) instead of the TxE command.

#### 6.9.1.4.2 EUTs not transmitting initial bursts

This applies only to EUTs that do not transmit initial bursts (i.e. where only "Initial phase-Standby" is implemented):

a) The EUT shall be set in "Initial phase-Standby" EST state.

b) The STE shall not send the TxE command to maintain the EUT in "Initial phase-Standby" EST state.

c) A condition requiring the cessation of emission shall be initiated either by the STE or the EUT.

d) In order to verify that the EUT remains in the "Initial phase-Standby" EST state, it shall be verified that the EUT remains in the "Emissions disabled" radio state when the EUT is requested to transmit data.

e) The STE shall send a TxE command to the EUT.

f) In order to verify that the EUT remains in the "Initial phase-Standby" EST state or enters the "Transmission disabled" EST state, it shall be verified that the EUT remains in the "Emissions disabled" radio state when the EUT is requested to transmit data.

g) Repeat the test steps a) through f) using a TxD command in step e) instead of the TxE command.

### 6.9.1.5 "Single action" means of cessation of emissions

a) It shall be verified that there is a "single action" means of ceasing emissions of the EUT (e.g. the switch thrown).

b) The EUT shall be set in the "Transmission enabled" EST state.

c) The EUT shall be set in the "Carrier on" radio state.

d) The "single action" means of cessation of emissions shall be activated.

e) It shall be verified that the EUT enters the "Emissions disabled" radio state.

f) In order to verify that the EUT is in the "Transmission disabled" EST state, it shall be verified that the EUT remains in the "Emissions disabled" radio state when the EUT is requested to transmit data.

g) The STE shall send a TxE command to the EUT.

h) In order to verify that the EUT remains in the "Transmission disabled" EST state, it shall be verified that the EUT remains in the "Emissions disabled" radio state when the EUT is requested to transmit data.

i) The EUT shall be switched off and then switched on and the EUT shall enter the "Initial phase" EST state without receiving a TxE from the STE.

j) The "single action" means of cessation of emissions shall be activated.

k) The STE shall send a TxE command to the EUT.

l) In order to verify that the EUT remains in the "Initial phase" EST state or enters the "Transmission disabled" EST state, it shall be verified that the EUT remains in the "Emissions disabled" radio state when the EUT is requested to transmit data.

m) The EUT shall be switched off and then switched on and the EUT shall enter the "Initial phase" EST state without receiving a TxE from the STE.

n) The STE shall send a TxD command to the EUT so that it enters the "Transmission disabled" EST state.

o) The "single action" means of cessation of emissions shall be activated.

p) The STE shall send a TxE command to the EUT.

q) In order to verify that the EUT remains in the "Transmission disabled" EST state, it shall be verified that the EUT remains in the "Emissions disabled" radio state when the EUT is requested to transmit data.

#### 6.9.1.6 Fault conditions

The EUT documentation shall include the means of generating fault conditions:

a) The EUT shall be set in the "Transmission enabled" EST state.

b) The EUT shall be set in the "Carrier on" radio state.

c) A fault condition requiring the cessation of emission shall be generated either by the STE or directly on the EUT.

d) It shall be verified that the EUT enters the "Emissions disabled" radio state as long as the fault condition persists.

e) Steps a) through d) shall be repeated for each fault condition defined for the intended use of the EST.

## 6.10 Identification of EST

### 6.10.1 Test arrangement

The test shall be arranged such that EUT can determine its own location, or if external means are employed, the test shall be arranged so that the EUT can be located by the means defined for the intended use of the EST. The STE shall be arranged to simulate the NCF. The actual location of the EME shall be known within the resolution required in clause 4.2.7.2.

### 6.10.2 Test method

a) The EUT shall be in the "Transmission enabled" state.

b) The EUT shall report its location to the STE.

c) The difference between the reported EUT location and actual location of the EUT (as simulated in the test) shall be within twice the resolution required in clause 4.2.7.2.

## 6.11 Control and monitoring functions

### 6.11.0 General

The EST can be modified for the purpose of these tests provided that full documentation is given to prove that the modifications accurately simulate the required test conditions.

The EUT shall transmit at $EIRP_{max}$.

For the purpose of these tests the EUT is the EST either with or without its antenna connected.

The measurement of the e.i.r.p. spectral density shall be limited within either the nominated bandwidth or to a 10 MHz bandwidth centred on the carrier frequency, whichever is the greater.

### 6.11.1 Test arrangement

The EUT shall be authorized to transmit and shall be in the "Transmission enabled" state at the commencement of each test, unless otherwise stated. The time difference between the command, or failure, and the occurrence of the expected event (e.g. the transmission suppression) shall be measured and recorded.

### 6.11.2 Processor monitoring- Test method

a) Each of the processors within the EUT shall, in turn, be caused to fail.

b) Within 1 second of such failure the EUT shall cease to transmit.

c) It shall be confirmed that the transmissions have been suppressed within the limits of the "Carrier-off" radio state emission levels.

d) Within 30 seconds of such failure it shall be confirmed that the transmissions have been suppressed within the limits of the "Emissions disabled" radio state emission levels.

e) The failed processor shall be restored to normal working condition and the EUT shall restore automatically to normal working before the next processor shall be induced to fail.

### 6.11.3 Transmit subsystem monitoring-Test method

a) The frequency generation subsystem within the EUT shall be caused to fail in respect of:

- loss of frequency lock (if applicable);
- absence of Local Oscillator (LO) output signal.

b) Recognition of each failure in turn by the subsystem monitor shall constitute an SMF event.

c) Within 1 second of such failure the EUT shall cease to transmit.

d) It shall be observed that the transmissions have been suppressed within the limits of the "Emissions disabled" radio state emission levels.

e) The failed elements shall be restored to normal working state and the EUT shall be restored to normal working condition before the next induced failure.

### 6.11.4 Power-on/Reset-Test method

a) The EUT shall be switched off and the STE shall not transmit the CC.

b) The EUT shall be switched on.

c) The EUT shall not transmit during and after switching-on, and shall enter the "Non valid" state.

If a manual reset function is implemented the following test shall be performed:

d) The EUT shall be restored to the "Initial phase" state and the STE shall transmit the CC.

e) The EUT shall remain in the "Initial phase" state.

f) The reset function shall be initiated.

g) The EUT shall enter the "Non valid" state.

h) The EUT shall be restored to the "Initial phase" state and the STE shall transmit the CC as well as a TxE.

i) The EUT shall enter the "Transmission enabled" state.

j) The reset function shall be initiated.

k) The EUT shall enter the "Non valid" state.

### 6.11.5 Control Channel and Response Channel -Test method

For the purposes of these tests, the period without correct reception of the CC (T1) shall be as specified in clause 4.2.8.5.2.2.

These test procedures require simulation of the following events:

a) The CC has never been received by the EUT after power-on.

b) The CC is lost by the EUT after reception of a Transmission Enable command.

c) The CC is lost by the EUT without reception of a Transmission Enable command.

d) The CC is being lost by the EUT and a call is initiated within the period T1.

a) Case where the CC has never been received by the EUT after power-on:

a1) the EUT shall be switched off and the STE shall not transmit the CC;

a2) the EUT shall be switched on;

a3) the EUT shall remain in the "Non valid" state.

b) Case where the CC is lost by the EUT after reception of a Transmission Enable command:

b1) the EUT shall be switched-on and the STE shall transmit the CC and a Transmission Enable command;

b2) the EUT shall enter the "Initial phase" state and go, if applicable, to the "Transmission enabled" state;

b3) a transmission request shall be initiated from the EUT;

b4) the STE shall stop transmitting the CC;

b5) within the period T1 from event b4), the EUT shall enter the "Non valid" state.

c) Case where the CC is lost by the EUT without reception of a Transmission Enable command:

c1) the EUT shall be switched on and the STE shall transmit the CC;

c2) the EUT shall enter the "Initial phase" state;

c3) the STE shall stop transmitting the CC;

c4) the EUT shall enter in the "Non valid" state not later than T1;

c5) a transmission request shall be initiated and the EUT shall remain in the "Non valid" state.

d) Case where the CC is being lost by the EUT and a call is initiated within the T1 period:

d1) the EUT shall be switched on and the STE shall transmit the CC;

d2) the STE shall stop transmitting the CC;

d3) within the period T1 from d2), a transmission request shall be initiated from the EUT;

d4) the EUT may transmit but within the T1 period the EUT shall enter the "Non valid" state.

### 6.11.6 Network Control commands-Test method

The tests shall be performed in the following sequence:

- Transmission Enable command;
- Transmission Disable command received in the "Transmission enabled" state;
- Transmission Disable command received in the "Initial phase" state.

a) Transmission Enable command:

a1) the EUT shall be switched-on and the STE shall transmit the CC;

a2) the EUT shall enter the "Initial phase" state;

a3) a transmission request shall be initiated from the EUT, the EUT shall remain in the "Initial phase" state;

a4) the STE shall transmit a transmit enable command to the EUT;

a5) a transmission request shall be initiated from the EUT;

a6) the EUT shall enter the "Transmission enabled" state and shall transmit.

b) Transmission Disable command received in the "Transmission enabled" state:

b1) continue from a6);

b2) the STE shall transmit a disable command to the EUT;

b3) the EUT shall enter the "Transmission disabled" state within 1 second;

b4) a transmission request shall be initiated from the EUT;

b5) the EUT shall remain in the "Transmission disabled" state;

b6) the STE shall transmit an enable command;

b7) the EUT shall enter either the "Transmission enabled" state or the "Initial phase" state;

b8) if the EUT is in the "Transmission enable" state then the test continues with b11);

b9) the STE shall transmit a TxE command;

b10) the EUT shall enter the "Transmission enable" state;

b11) if a transmission request is not active any more than a new transmission request shall be initiated;

b12) the EUT shall transmit;

b13) the EUT transmission shall be terminated.

c) Transmission Disable command received in the "Initial phase" state:

c1) the EUT shall be switched-on and the STE shall transmit the CC;

c2) the EUT shall enter the "Initial phase" state;

c3) the STE shall transmit a Transmission Disable command to the EUT;

c4) the EUT shall enter the "Transmission disabled" state within 1 second;

c5) a transmission request shall be initiated from the EUT;

c6) the EUT shall remain in the "Transmission disabled" state;

c7) the STE shall transmit a Transmission Enable command;

c8) the EUT shall enter either the "Transmission enabled" state or the "Initial phase" state;

c9) if the EUT is in the "Transmission enable" state then the test continues with c12);

c10) the STE shall transmit a TxE command;

c11) the EUT shall enter the "Transmission enable" state;

c12) if a transmission request is not active any more than a new transmission request shall be initiated;

c13) the EUT shall transmit;

c14) the EUT transmission shall be terminated.

### 6.11.7 Initial burst transmission-Test method

The EUT documentation shall include any means to identify the initial burst duration.

The following test shall be performed in sequence:

a) the EUT shall be switched-off and the STE shall transmit the CC;

b) the EUT shall be switched-on;

c) the EUT shall not transmit, except the initial bursts;

d) it shall be verified that the specifications given in clause 4.2.8.7.2 are fulfilled.

### 6.11.8 Inhibition of transmission-Test method

The EUT documentation shall include the values for $IT_{max}$ and $IT_{trans}$, as specified in clause 4.2.8.8.

The receive carrier to the EUT shall be removed:

a) Within 0,5 seconds of such failure the EUT shall cease to transmit.

b) It shall be observed that the EUT remains in the "Transmission enabled" state and that transmissions have been suppressed within the limits of the "Carrier off" radio state emission levels given in clause 4.2.1.2, table 3.

c) The receive carrier to the EUT shall be restored within $IT_{max}$.

d) It shall be observed that the EUT enters the "Carrier on" radio state and begins transmission within $IT_{trans}$.

e) The receive carrier to the EUT shall be removed.

f) It shall be observed that after $IT_{max}$ the EUT enters the "Initial phase" state.

## 6.12 Receive antenna off-axis gain pattern

### 6.12.1 Test Method

#### 6.12.1.1 Test site

This test shall be performed using the method of measurement and test arrangement as specified in the following clauses.

The tests can be carried out in an Open Area Test Site (OATS) environment or in an anechoic chamber or in a semi-anechoic chamber as they lead to comparable test results for establishing compliance. Ambient noise levels shall be at least 6 dB below the applicable unwanted emissions limit. The preferred test site is the semi-anechoic chamber.

#### 6.12.1.2 Method of measurement

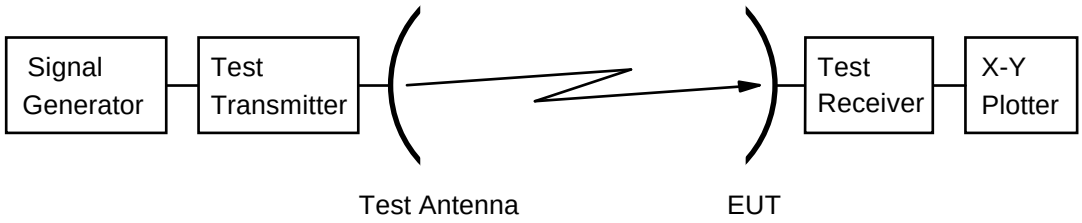



**Figure 9: Test arrangement - antenna receive pattern measurement**

a) The test arrangement shall be as shown in figure 9 with the EUT connected to the test receiver.

b) A signal proportional to the angular position from the servo mechanism shall be applied to the X-axis and the signal level from the test receiver shall be applied to the Y-axis of the plotter or alternatively in the computerized case.

c) The test frequencies shall be the centre frequency of each applicable frequency range. The E plane shall be vertical.

d) The EUT shall be aligned to maximize the received signal level and the X-Y plotter, or alternatively in the computerized case, shall be adjusted to give the maximum reading on the chart.

e) The EUT shall be driven in azimuth through 180°.

f) The pattern measurement is then obtained by driving the EUT in azimuth through 360° with the set up recording the results.

g) The tests in b) to e) shall be repeated with the frequency changed to the lower limit of the applicable band as defined for the intended use of the EST.

h) The tests in b) to e) shall be repeated with the frequency changed to the upper limit of the applicable band as defined for the intended use of the EST.

i) The tests in b) to h) shall be repeated with the frequencies changed to the others specified if the design of the equipment is such that operation is possible, but not necessarily simultaneously, in all bands.

j) The tests in b) to h) shall be repeated with the test signal being transmitted in the H-plane instead of the E-plane.

k) The tests in b) to h) shall be repeated with the test signal being transmitted in a plane at 45° to the H-plane.

l) The tests in b) to h) shall be repeated with the test signal being transmitted in a plane at 90° to that in k).

m) The tests in b) to l) shall be repeated between the angles of $\varphi_r$ and 7° with the EUT rotated through 90°, or the test antenna or the polarization subsystem of the EUT rotated by 90°, to give the cross-polar measurement.

## 6.13 Blocking performance

### 6.13.1 Test method

a) The output signals of two signal generators shall be combined with equal weight. The first signal shall be the wanted modulated signal as defined for the intended use, tuned at the centre frequency of the receive frequency band and have a level in accordance with the defined intended use. The second signal shall be a CW signal with same level. The combined signal shall be coupled to the LNB input.

b) A spectrum analyser shall be connected to the LNB output in a way that allows to supply the LNB with power.

c) The first signal generator frequency shall be set to $f_c$. $f_c$ is the center frequency of the receive frequency band.

d) The first signal generator level shall be set to a level in the LNB operational input level range.

e) The spectrum analyser shall be set for measuring the level of the converted first signal at the LNB output.

f) The second signal generator frequency shall be set to $f_c - 20$ MHz.

g) The second signal generator level shall be adjusted so that the measured level is 1 dB less than in absence of the second signal.

h) The second signal generator level shall be noted down as reference level.

i) The second signal generator frequency shall be set to the frequency of interest.

j) The second signal generator level shall be adjusted so that the measured level is 1 dB less than in absence of the second signal.

k) Steps i) to j) shall be repeated for frequencies in the ranges of table 4. The rejection at the frequency of interest is equal to the second signal generator level minus the reference level determined in step i).

NOTE: The worst case rejection in a particular frequency range can be determined after step i) by sweeping the second signal generator frequency over the frequency range and observe the gain compression, then perform steps j) to k) with the frequency where gain compression is highest.

# 6.14 Adjacent Signal Selectivity

## 6.14.1 Test method

a) Two test signal generators shall be used each generating the same modulated signal as the wanted signal in the IME input frequency range and thermal noise as defined for the intended use.

b) The signal generators shall be connected to the IME input through a splitter (combiner).

c) The test signal generators shall be set to the frequencies and levels according to table 5.

d) The IME shall be set to receive the signal of the first test signal generator.

e) The second test signal generator shall be set to signal off.

f) The noise level (or signal to noise ratio) of the first test signal generator shall be varied in order to determine the threshold reference sensitivity level, such that the SNR (at the receiver input) is set to the reference level plus 1 dB. The reference level is the minimum SNR required to achieve the target performance level for the modulated wanted signal This is the sensitivity reference level.

g) The second signal generator shall be set to signal on.

h) The noise level (or signal to noise ratio) of the first test signal generator shall be varied in order to determine the threshold reference sensitivity level.

i) The degradation is equal to the noise level (or signal to noise ratio) determined in step g) minus the signal value recorded in step f).

The result is the highest degradation found.

# Annex A (informative): Relationship between the present document and the essential requirements of Directive 2014/53/EU

The present document has been prepared under the Commission's standardisation request C(2015) 5376 final [i.5] to provide one voluntary means of conforming to the essential requirements of Directive 2014/53/EU on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment and repealing Directive 1999/5/EC [i.12].

Once the present document is cited in the Official Journal of the European Union under that Directive, compliance with the normative clauses of the present document given in table A.1 confers, within the limits of the scope of the present document, a presumption of conformity with the corresponding essential requirements of that Directive, and associated EFTA regulations.

**Table A.1: Relationship between the present document and the essential requirements of Directive 2014/53/EU**

| Harmonised Standard ETSI EN 302 448 | | | | | |
|---|---|---|---|---|---|
| Requirement | | | | Requirement Conditionality | |
| No | Description | Essential requirements of Directive | Clause(s) of the present document | U/C | Condition |
| 1 | Off-axis spurious radiation | 3.2 | 4.2.1 | U | |
| 2 | On-axis spurious radiation | 3.2 | 4.2.2 | U | |
| 3 | Off-axis e.i.r.p. emission density within the band | 3.2 | 4.2.3 | U | |
| 4 | Carrier suppression | 3.2 | 4.2.4 | U | |
| 5 | Antenna pointing | 3.2 | 4.2.5 | U | |
| 6 | Cessation of emissions of the EST | 3.2 | 4.2.6 | U | |
| 7 | Identification of EST | 3.2 | 4.2.7 | U | |
| 8 | Control and Monitoring Functions (CMFs) | 3.2 | 4.2.8 | U | |
| 9 | Antenna mechanical stability | 3.2 | 4.2.5 | U | |
| 10 | Antenna polarization alignment | 3.2 | 4.2.5 | U | |
| 11 | Receive antenna off-axis gain pattern | 3.2 | 4.2.9 | C | Satellite communications networks may contain a central control unit that keeps received interference low by taking into account the actual antenna gain pattern of earth stations. The requirement is not relevant for earth stations that are always operated as part of such a network. The requirement is relevant in all other cases.<br>If the receive antenna performance does not meet the requirement of clause 4.2.9, then the earth station can accept an additional co-frequency interference caused by its off-axis gain which is not compliant to the mask<br>This requirement is not required for Earth Stations always operated as part of a network with a central control unit. |
| 12 | Blocking performance | 3.2 | 4.2.10 | U | |
| 13 | Adjacent Signal Selectivity | 3.2 | 4.2.11 | U | |

| Harmonised Standard ETSI EN 302 448 | | | | | |
|---|---|---|---|---|---|
| Requirement | | | | Requirement Conditionality | |
| No | Description | Essential requirements of Directive | Clause(s) of the present document | U/C | Condition |
| 14 | Receiver radio-frequency intermodulation | 3.2 | | | In an FSS network, receiver radio-frequency intermodulation is not relevant, because inter-modulating signals originating from other networks are rejected thanks to requirements on receive antenna off-axis gain pattern and adjacent signal selectivity. Interfering signals up to 28 dB above the wanted carrier can be tolerated with minimal impact on performance. |
| 15 | Receiver sensitivity | 3.2 | | | There is no causal relationship between receiver sensitivity and interference in the case of satellite communications. Therefore, a quantitative calculation is not possible. |
| 16 | Receiver spurious response rejection | 3.2 | | | Frequencies other than the image frequency are not relevant because this is covered by receiver blocking and adjacent channel requirements which are already included in the receiver parameters. |
| 17 | Receiver multiple signal selectivity - receiver adjacent signal selectivity | 3.2 | | | The proper function of the wanted signal in the presence of other unwanted emissions is controlled via proper assignment of frequencies and frequency coordination procedures as applicable. Due to the operation without protection against interference, there is no need of specifying this parameter. |
| 18 | Receiver dynamic range | 3.2 | | | In an FSS network, receiver dynamic range has no influence on harmful interference. The satellite earth stations are deployed in a manner that results in a situation where it is impossible to receive a wanted signal that is high enough to produce any overloading effect (satellite systems are power limited). So that it is not necessary to specify receiver dynamic range. See clause 5.3.6.1 of ETSI EG 203 336 [i.13]. |
| 19 | Reciprocal mixing | 3.2 | | | For FSS equipment, the parameter is excluded for the following reason. It is considered that the reciprocal mixing effects are implicitly covered in HSs where comprehensive interference characteristics are specified in terms of selectivity and/or blocking requirements, thus removing the need for this parameter to be included in HSs as the effects of receiver selectivity and reciprocal mixing cannot be separated. See clause 5.3.6.2 of ETSI EG 203 336 [i.13]. |

| Harmonised Standard ETSI EN 302 448 | | | | | |
|---|---|---|---|---|---|
| Requirement | | | | Requirement Conditionality | |
| No | Description | Essential requirements of Directive | Clause(s) of the present document | U/C | Condition |
| 20 | Desensitization | 3.2 | | | As desensitization is a receiver effect addressed by other parameters (including receiver blocking), its inclusion as a separate parameter in an HS is not required. See clause 5.3.2.3 of ETSI EG 203 336 [i.13]. |
| 21 | Receiver unwanted emissions in the spurious domain | 3.2 | | | Frequencies other than the image frequency are not relevant because this is covered by receiver blocking and adjacent channel requirements which are already included in the receiver parameters. |

**Key to columns:**

**Requirement:**

**No** A unique identifier for one row of the table which may be used to identify a requirement.

**Description** A textual reference to the requirement.

**Essential requirements of Directive**

Identification of article(s) defining the requirement in the Directive.

**Clause(s) of the present document**

Identification of clause(s) defining the requirement in the present document unless another document is referenced explicitly.

**Requirement Conditionality:**

**U/C** Indicates whether the requirement is unconditionally applicable (U) or is conditional upon the manufacturer's claimed functionality of the equipment (C).

**Condition** Explains the conditions when the requirement is or is not applicable for a requirement which is classified "conditional".

Presumption of conformity stays valid only as long as a reference to the present document is maintained in the list published in the Official Journal of the European Union. Users of the present document should consult frequently the latest list published in the Official Journal of the European Union.

Other Union legislation may be applicable to the product(s) falling within the scope of the present document.

# Annex B (normative): Mechanical stability methodology

This test based on an analytical or numerical analysis is performed in two stages.

In the first stage the loads generated by the maximum wind speed should be computed on the EME, taking into account the intrinsic properties of the materials.

In the second stage the computed loads is applied to the structure.

The purpose of the analysis is twofold:

a) To show that the torque and the fields of force applied to the EME structure under nominated conditions do not reach any of the following:

- the point where deformation fouls the antenna pointing mechanism;
- the yield point of any element of the structure;
- the buckling point of any element of the structure.

NOTE: The yield point is elastic limit of the material and the buckling point is the failure point for the material.

b) To compute equivalent static loads (force and torque) applied to the critical attachment points of the structures, e.g.:

- for antennas without a radome:
  - reflector-mounting legs fixing point;
  - reflector-struts;
  - LNB- struts;
- for antennas with a radome:
  - radome surface;
  - radome mounting points.

Analysis and load applications procedure:

a) The air related parameters, namely the kinetic viscosity used to calculate drags at the rims of the structure should be calculated with the standard atmospheric environmental conditions (temperature = 293 K, air pressure = $1{,}013 \times 10^5$ Pascal).

b) The computations needed to derive the field of force and torque and the equivalent static stresses should be carried out for each of the following variables:

- elevation angle: maximum and minimum for antennas without a radome;
- wind direction: in steps of 45° around the EME unit;
- maximum relative wind speed: 100 km/h;
- gust wind speed: 130 km/h for 3 seconds.

It shall be verified with the simulated results that yield point limits are not exceeded for any self-contained element.

The calculated equivalent static loads is applied at any identified critical fixing point of the assembly, whilst the loads are applied the above deck unit should be observed and any distortion noted.

The test report shall contain the following information:

a) the computation method used;

b) the description of the test equipment;

c) the description of the tests performed;

d) the results of the safety margin test;

e) any signs of distortion observed;

f) the results of the measurements of the deviation of the antenna or radome position;

g) the component deviation with respect to each other.

# Annex C (informative): Maximum measurement uncertainty

The measurements described in the present document are based on the following assumptions:

- the measured value related to the corresponding limit is used to decide whether an equipment meets the requirements of the present document;
- the value of the measurement uncertainty for the measurement of each parameter is included in the test report.

Tables C.1 and C.2 show the recommended values for the maximum measurement uncertainty figures.

**Table C.1: Maximum measurement uncertainty**

| Parameter | Uncertainty |
|---|---|
| Radio frequency | ±10 kHz |
| RF power | ±0,75 dB |
| Conducted spurious | ±4 dB |
| Radiated spurious | ±6 dB |
| Antenna on-axis gain | ±2 dB |

**Table C.2: Maximum measurement uncertainties for antenna gain pattern**

| Gain relative to the antenna on-axis gain | Uncertainty |
|---|---|
| > -3 dB | ±0,3 dB |
| -3 dB to -20 dB | ±1,0 dB |
| -20 dB to -30 dB | ±2,0 dB |
| -30 dB to -40 dB | ±3,0 dB |

# Annex D (informative): Bibliography


- ETSI ETR 169 (1995): "Satellite Earth Stations and Systems (SES); Common Technical Regulations (CTRs) in the satellite earth station equipment field".
- EN 50121-3-2 (2019): "Railway applications - Electromagnetic compatibility - Part 3-2: Rolling stock - Apparatus", (produced by CEN).
- EN 50121-3-1 (2019): "Railway applications - Electromagnetic compatibility - Part 3-1: Rolling stock - Train and complete vehicle", (produced by CEN).
- CEPT/ERC/Recommendation 74-01 (2022): "Unwanted emissions in the spurious domain".

# Annex E (informative): Change history

| Version | Information about changes |
|---|---|
| V2.1.2 | Editorials suggested by WG HARM#83 WG HARM#85 on references to "intended use of the EST" from previously approved V1.1.5 and brought into the published version 2.1.1 |
| V2.1.3 | Review by WG HARM#88 |
| V2.1.4 | Clean version after review by WG HARM#88 |
| V2.1.5 | Further revision addressing ETSI Comments |
| V2.1.6 | Complete revision addressing ETSI Comments by WG HARM 99 |
| V2.1.7 to 2.1.10 | Intermediate reviews the HAS comments by WG HARM |
| V2.1.11 | Full review of HAS comments and adding missing test procedures for some parameters, aligning with EN 302 340, by WG HARM |
| V2.1.12 | Rapporteur formatting edits and updates reference year for some reference documents. Clean version for approval by TC SES. Approved by TC SES 109 |
| V2.1.13 | Minor editorial updates by WG HARM 136 to align with other ongoing HAS comments revisions to other ENs |
| V2.1.14 | Minor editorial updates by WG HARM 137 to align with other ongoing HAS comments revisions to other ENs |
| V2.1.15 | Minor editorial updates by WG HARM 138 to align with other ongoing HAS comments revisions to other ENs |
| V2.1.16 | Addition clause 6.8 as void which was unintentional omitted since version 2.1.15 and clarification of the step a) of 6.13 to address the modulated signal levels aligned with EN 301 428 and addressing similar HAS requirements as for other EN like EN 302 977, edited by WG HARM 139 |
| V2.1.17 | Review by WG HARM 148 to update references to OATS |

# History

| Version | Date | Status |
|---|---|---|
| V1.1.1 | December 2007 | Publication |
| V2.1.1 | May 2016 | Publication |
| V2.1.18 | September 2026 | SRdAP process EV 20261210: 2026-09-17 to 2026-12-10 |
| | | |
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